

1. Project Title: Real-Time Taxi Ride Analytics Pipeline

2. Type of System: Streaming Data Engineering Pipeline

-> The project simulates how a company processes real-time transportation data and generates analytics such as:

- ride counts
- average fares
- busiest pickup zones
- ride demand trends

Instead of processing data in batches, the system processes taxi ride events continuously as they arrive.

3. Logical Problem Statement:

-> Many modern systems (ride-sharing apps, financial transactions, IoT devices) generate continuous streams of data.

-> Traditional batch systems process data after it is stored, which causes delays.

-> Goal of the project:- Build a pipeline that can:

- ingest streaming data
- process it in real time
- simulate real-world streaming systems

-> Example use case: Taxi companies want to analyze ride data as it happens to monitor:

- ride demand
- location trends
- trip duration
- revenue patterns

4. Project Problem Statement:

-> Urban transportation systems generate massive streams of ride data every second.

-> Traditional batch processing systems analyze data hours later, which is not suitable for real-time decision making.

-> Example problems:

- city authorities cannot monitor ride demand in real time
- taxi companies cannot detect busy zones immediately
- analytics dashboards update too slowly.

-> Therefore, we need a real-time data pipeline that can:

- ingest streaming ride events
- process them instantly
- store metrics for analysis
- enable real-time insights

5. Aim of the Project

The aim is to design and implement a real-time data engineering pipeline that:

1. Simulates taxi ride events from the NYC Taxi dataset
2. Streams the data using Kafka
3. Processes the stream using Spark
4. Stores analytics metrics in Cassandra
5. Uses Airflow to orchestrate and manage the pipeline

This project demonstrates core data engineering concepts:

- streaming ingestion
- distributed processing
- scalable storage
- workflow orchestration

6. Dataset:

-> The data is taken from the **NYC Taxi dataset** from the New York City Taxi and Limousine Commission.

-> The dataset contains:

- pickup location
- dropoff location
- passenger count
- trip distance
- fare amount
- Timestamps

We simulate this historical dataset as live streaming events. Instead of sending the whole dataset at once, we send rows one by one with a delay.

7. Project Overview:

-> This project builds a real-time data engineering pipeline that processes taxi ride events and produces analytics that can be visualized on a dashboard. The dataset used is the NYC Taxi Trip dataset, which contains information about taxi rides such as pickup time, drop-off time, distance, fare amount, etc. Instead of processing the dataset as a static batch, the project simulates real-time streaming of taxi rides and processes them continuously to generate insights.

8. Goal of the Project:

-> The aim is to demonstrate how modern data engineering systems handle streaming data pipelines.

-> The pipeline will:

1. Ingest ride events
2. Stream them through a messaging system
3. Process them in real time
4. Store analytics metrics
5. Visualize insights on a dashboard

9. Industrial tools being used:

- Python
- Apache Kafka
- Apache Spark
- Apache Cassandra
- Apache Airflow
- Docker
- Grafana/Power BI

10. Architecture of the Pipeline:

-> The system follows this architecture:



11. Explanation of Each Stage:

1. Dataset

- Source data containing historical taxi ride records.

2. Python Producer

- Reads rows from the dataset.
- Converts them to JSON messages.
- Sends them to Kafka one event at a time to simulate real-time streaming.

3. Python: Writing pipeline logic

- Simulate taxi ride events
- Send data to kafka
- Write spark streaming jobs
- Interact with Cassandra
- Write Airflow DAGs
- Used because it works good with spark, karka, airflow

4. Kafka: Real-time streaming platform / real time data ingestion

- Kafka acts as a distributed event streaming platform.
- Acts as a message broker (data pipeline highway)
- Stores incoming ride events in a topic called taxi_rides.
- Instead of sending data directly to Spark, we push it into Kafka first.
- Kafka stores data as topics. Every taxi ride becomes a message.
- Allows downstream systems to consume the stream.

- Used as it is the industry standard of event pipelines, handles high throughput streaming, decouples producers and consumers, scalable and fault-tolerant, also used by netflix and uber.

5. Zookeeper: Kafka cluster coordination

- ZooKeeper was used in Kafka to manage broker metadata, leader election, cluster coordination, and topic configuration. Modern Kafka versions are moving to KRaft mode to remove the ZooKeeper dependency.
- ZooKeeper manages broker metadata, leader election, cluster synchronization

6. Spark Streaming: Real time data processing

- Spark Structured Streaming reads events from Kafka and processes them in micro-batches.
- Read messages from Kafka in real time.
- Processes the ride events.
- Computes analytics metrics such as trip counts, averages, or revenue trends.
- Supports large scale data processing, structured streaming, and machine learning

7. Data Lakehouse Architecture : Bronze–Silver–Gold Layers

Bronze, Silver, and Gold layers represent different levels of data processing in modern data engineering systems.

-> Bronze Layer – Raw Data

- Stores raw taxi ride events exactly as received from Kafka.
- No transformation is applied.
- Acts as the raw historical record of the data stream.

->Silver Layer – Cleaned Data

- Raw events are cleaned and validated.
- Removes null values, invalid distances, or corrupt records.
- Converts timestamps into structured formats.

->Gold Layer – Aggregated Analytics

- Business-level analytics are generated.
- Example metrics:
 - average fare
 - average trip distance
 - ride counts
 - revenue trends
- These aggregated metrics are stored in Cassandra for fast querying.

This architecture is commonly used in modern data platforms such as Databricks Lakehouse architectures.

Bronze and Gold layers are persisted in Cassandra.

Silver is an intermediate transformation layer in Spark

8. Cassandra: Data storage for analytics

- Storage layer for processed metrics
- Spark writes different levels of processed data into the system: Bronze Layer – raw ride events | Silver Layer – cleaned and validated ride data | Gold Layer – aggregated

analytics metrics. Cassandra stores the final Gold layer analytics results so that they can be queried efficiently by dashboards such as Power BI or Grafana.

- Stores processed analytics results.
- Optimized for fast reads and writes for streaming workloads.
- Cassandra is used for High speed, Good for streaming data, and has a Distributed NoSQL database.

9. Apache Airflow: Pipeline Orchestration

- Airflow schedules and manages workflows.
- Instead of manually running scripts, Airflow runs them automatically.
- Airflow uses DAGs (Directed Acyclic Graphs).
- Airflow is used to automate pipelines, schedule jobs, monitor pipelines, and is the industry standard.

10. Docker: Infrastructure management

- Run all tools easily
- Installing Kafka, Spark, Cassandra manually is difficult.
- Docker lets us run them as containers.
- Benefits include easy steps, no dependency conflicts, same environment everywhere.

11. Power BI

- Connects to the processed data.
- Displays analytics through dashboards and visualizations.

STEP I

1. Made the project folder 'taxistream'
2. Inside the folder taxistream, made the folders:

```
>airflow
>data
>docker
>producer
  >producer.py
>spark
  >stream_processor.py
>docker-compose.yml
>Readme.md
>test_data.py
>requirements.txt
```

STEP II: ENVIRONMENT SETUP

1. In Anaconda Terminal we create a new environment:

```
>conda create -n taxic python=3.10
>conda activate taxic
```
2. We install Basic libraries:

```
>pip install kafka-python pyspark cassandra-driver pandas pyarrow apache-airflow
```

This will be used for Python Producer → Kafka, Spark Streaming, Cassandra connection.
 - Pyarrow is used for .parquet files.

- kafka-python is used to connect Python producer to Kafka, and send taxi events to Kafka topics.
 - pyspark is used to implement Spark streaming jobs, and process data from Kafka
 - pandas is used to read and preprocess the NYC dataset, and simulate streaming data
 - cassandra-driver is used to connect Spark/Python to Cassandra database, and insert processed analytics
 - apache-airflow is used to define workflow DAGs, and schedule pipeline tasks
3. We save the requirements in the taxistream folder as a txt file:
- ```
>conda list >requirements.txt
```

### STEP III: DATASET DOWNLOAD

1. Go to the link: <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>  
(This dataset has yellow taxi trip records, green taxi trip records, for each month and year)
2. Download the "2025, yellow taxi trip records (parquet)" and put that inside >data folder.
3. So now the >data folder has >yellow\_tripdata\_2025-01.parquet.  
(.parquet is a columnar data format used in big data systems like Spark, hive, hadoop, so it actually makes the project look realistic)

.....

Testing:

1. We open the terminal:
 

```
>activate taxic
>cd "C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream"
```
2. We made the >test\_data.py under the taxistream folder by using the terminal command:
 

```
>type nul > test_data.py
```
3. Inside it, we write the code:
 

```
import pandas as pd
df = pd.read_parquet("data/yellow_tripdata_2025-01.parquet")
print(df.head())
print("Total rows:", len(df))
```
4. We run this code in the terminal:
 

```
>python test_data.py
```

```
VendorID tpep_pickup_datetime tpep_dropoff_datetime ... congestion_surcharge Airport_fee cbd_congestion_fee
0 1 2025-01-01 00:18:38 2025-01-01 00:26:59 ... 2.5 0.0 0.0
1 1 2025-01-01 00:32:40 2025-01-01 00:35:13 ... 2.5 0.0 0.0
2 1 2025-01-01 00:44:04 2025-01-01 00:46:01 ... 2.5 0.0 0.0
3 2 2025-01-01 00:14:27 2025-01-01 00:20:01 ... 0.0 0.0 0.0
4 2 2025-01-01 00:21:34 2025-01-01 00:25:06 ... 0.0 0.0 0.0

[5 rows x 20 columns]
Total rows: 3475226
```

5. Milestone completed: Dataset -> Python

### STEP IV: DOCKER INFRASTRUCTURE

(To set up docker containers for heavy tools such as: Zookeeper | kafka | Spark | Cassandra | Airflow. All these will be controlled by Docker Compose)

1. We installed Docker Desktop, Windows ARM64 from the link :  
<https://www.docker.com/products/docker-desktop/> .

(Without Docker you would need to manually install and configure: Apache Kafka | Apache ZooKeeper | Apache Spark | Apache Cassandra | Apache Airflow. Each of those installations can take hours and break easily.

With Docker we run them using one command:

```
>docker compose up)
```

2. We can then check Docker version:

```
>docker --version
```

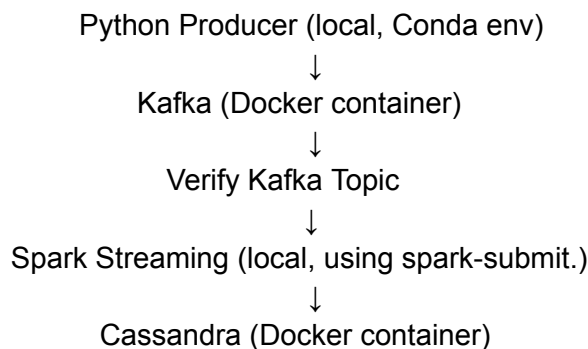
```
Docker version 20.10.1, build a5c7197
```

3. The >docker-compose.yml file will be used to start Apache Kafka and Apache Zookeeper inside docker containers. Hence, **Docker is the infrastructure layer** that runs Kafka, Spark, Cassandra, etc.

4. While the conda env manages python and its libraries(pandas, pyarrow, pyspark, kafka-spark), Docker on the other hand runs external services your Python code connects to, such as: Apache Kafka, Apache ZooKeeper, Apache Spark, Apache Cassandra, Apache Airflow.

### How they connect?:

Example pipeline:



**They communicate through ports like:** localhost:9092 → Kafka

So, in short, Python scripts use the tools to send and process data, while Docker runs the data infrastructure. Python and Docker work together using localhost connections.

5. After installing, we create the file >docker-compose.yml under the taxistream folder.

(This file will start the services your pipeline needs. For now we will start with Kafka infrastructure: Apache Kafka and Apache ZooKeeper.

Later we will add: Apache Cassandra, Apache Spark, Apache Airflow

6. In >docker-compose.yml, we write the code:

```
services:

 zookeeper:
 image: confluentinc/cp-zookeeper:7.5.0
 container_name: zookeeper
 environment:
 ZOOKEEPER_CLIENT_PORT: 2181
```

```

ports:
 - "2181:2181"

kafka:
 image: confluentinc/cp-kafka:7.5.0
 container_name: kafka
 depends_on:
 - zookeeper
 ports:
 - "9092:9092"
 environment:
 KAFKA_BROKER_ID: 1
 KAFKA_ZOOKEEPER_CONNECT: zookeeper:2181
 KAFKA_ADVERTISED_LISTENERS: PLAINTEXT://localhost:9092
 KAFKA_OFFSETS_TOPIC_REPLICATION_FACTOR: 1

```

7. In the Anaconda prompt, we execute:

```
>docker compose up
```

It will ask for : View in Docker Desktop | View Config | Enable Watch | Detach

We execute:

```
>docker compose up -d
```

Docker will download images for Apache ZooKeeper and Apache Kafka (because the code has it only for zookeeper and kafka)

-> We can check the containers by:

```
>docker ps
```

```

(taxic) C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream>docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
fc372b32f9c9 confluentinc/cp-kafka:7.5.0 "/etc/confluent/dock... 3 days ago Up 2 days 0.0.0.0:9092->9092/tc
p, [::]:9092->9092/tcp kafka
1b42226a5712 confluentinc/cp-zookeeper:7.5.0 "/etc/confluent/dock... 3 days ago Up 2 days 0.0.0.0:2181->2181/tc
p, [::]:2181->2181/tcp zookeeper

```

So, Apache ZooKeeper is running on port 2181

Apache Kafka is running on port 9092

So our streaming infrastructure is now alive.

### STEP V: PYTHON KAFKA PRODUCER

(Python Kafka producer that sends taxi rides from the dataset into Kafka.)

1. We create file 'producer.py' under >producer folder in the taxistream folder.

2. In producer.py, we run the code:

```

from kafka import KafkaProducer
import pandas as pd
import json
import time

Kafka connection

```

```

producer = KafkaProducer(
 bootstrap_servers='localhost:9092',
 value_serializer=lambda v: json.dumps(v).encode('utf-8')
)

Load dataset
df = pd.read_parquet('data/yellow_tripdata_2025-01.parquet')

Limit rows for testing
df = df.head(50)

topic = "taxi_rides"

for index, row in df.iterrows():

 data = row.to_dict()

 # Convert timestamps to string
 for key, value in data.items():
 if hasattr(value, "isoformat"):
 data[key] = value.isoformat()

 producer.send(topic, value=data)

 print(f"Sent ride {index}")

 time.sleep(0.5) # simulate real-time streaming

```

Then we run the command in anaconda prompt:

```
>python producer/producer.py
```

```

(taxic) C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream>python producer/producer.py
Sent ride 0
Sent ride 1
Sent ride 2
Sent ride 3
Sent ride 4
Sent ride 5
Sent ride 6
Sent ride 7
Sent ride 8
Sent ride 9
Sent ride 10
Sent ride 11
Sent ride 12
Sent ride 13
Sent ride 14
Sent ride 15

```

3. If for any reason, the keyline is : “NoBrokerAvailable”, that means that Kafka isn’t reachable at `localhost:9092` when the producer starts

What to do:

```
>docker compose down
>docker compose up -d ; and then,
>python producer/producer.py
```

Also, keep docker desktop on.

---

### BRIEF OF WHAT WE HAVE DONE TILL NOW:

1. We created a conda environment for the project, and installed libraries like: pandas, kafka-python, pyspark, and pyarrow. This environment is where our pipeline code runs.
2. Infrastructure with Docker: We installed Docker and started containers for: Apache ZooKeeper and Apache Kafka. These run as background services. Kafka is basically a real-time data stream platform.

#### Analogy:

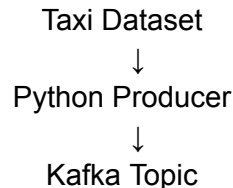
Think of it like a message pipeline where systems can:

send data → store temporarily → allow other systems to read it

3. Created a Kafka Producer: We wrote [producer.py](#), the program:

- A. Reads the NYC taxi dataset
- B. Converts each row to JSON
- C. Sends it to Apache Kafka

4. Hence, so far, the flow is:



5. We also added: `time.sleep(0.5)`

This **simulates real-time streaming** instead of sending everything instantly.

So our program sent:

Sent ride 0

Sent ride 1

...

Sent ride 49

These messages are now inside Kafka.

6. What Kafka is doing right now:

Kafka stores messages inside something called a topic.

Topic : `taxi_rides`

So internally it looks like:

Kafka Topic: `taxi_rides`

Ride 0

Ride 1

Ride 2

...

Kafka is waiting for another system to **consume** this stream.

7. To create a kafka topic under docker:

```
>docker exec -it kafka kafka-topics --create --topic taxi_rides --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1
```

8. **Before moving forward, check whether kafka is working or not:**

```
>docker exec -it kafka kafka-topics --list --bootstrap-server localhost:9092
```

```
(taxic) C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream>docker
exec -it kafka kafka-topics --list --bootstrap-server localhost:9092
taxi_rides
```

```
(taxic) C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream>
```

---

### BRIEF OF WHAT HAS TO BE DONE NOW:

1. Need to create a consumer from Apache Spark, which will:
  - Read Messages from Kafka
  - Process them in real time
  - Compute analytics metrics like: rides per minute, avg trip distance, avg fare amount, total revenue trends
2. Spark streaming consumer: Spark will continuously read the kafka stream, parse the incoming ride events, and perform real time processing. It will also compute metrics such as given above.  
Cassandra will store analytics results, so that they can be queried efficiently.  
Powerbi will visualize analytics such as: trip volume over time, revenue trends, avg trip details, peak ride hours.
3. Final Outcome: The completed system will function as a real-time taxi analytics platform, demonstrating how streaming data pipelines ingest, process, store, and visualize large volumes of event data.

---

### STEP VI: SPARK STREAMING SCRIPT

1. We create a file >stream\_processor.py under >spark folder in taxistream.

2. We write the code in stream\_processor.py:

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, from_json
from pyspark.sql.types import StructType, StringType, DoubleType

Create Spark session
spark = SparkSession.builder \
 .appName("TaxiStreamingProcessor") \
 .config("spark.sql.shuffle.partitions", "2") \
 .getOrCreate()
```

```

spark.sparkContext.setLogLevel("WARN")

Schema for taxi data
schema = StructType() \
 .add("tpep_pickup_datetime", StringType()) \
 .add("tpep_dropoff_datetime", StringType()) \
 .add("trip_distance", DoubleType()) \
 .add("fare_amount", DoubleType())

Read stream from Kafka
df = spark.readStream \
 .format("kafka") \
 .option("kafka.bootstrap.servers", "localhost:9092") \
 .option("subscribe", "taxi_rides") \
 .option("startingOffsets", "latest") \
 .load()

Convert Kafka value to string
json_df = df.selectExpr("CAST(value AS STRING)")

Parse JSON
taxi_df = json_df.select(
 from_json(col("value"), schema).alias("data")
).select("data.*")

Compute analytics
analytics_df = taxi_df.groupBy().avg("fare_amount", "trip_distance")

Output to console (for testing)
query = analytics_df.writeStream \
 .outputMode("complete") \
 .format("console") \
 .start()

query.awaitTermination()

```

3. We go to <https://adoptium.net/> , and install Java from there. Temurin 17.0.18+8, Windows 64 bit (.MS
4. We see that this java file has been installed here:

C:\Program Files\Eclipse Adoptium\jdk-17.0.18-hotspot  
Inside this, there should be a bin folder, having java.exe

### STEP VII: Setting JAVA\_HOME system variable

1. Press Windows Key → Search “Environment Variables” → Edit the system environment variables → Environment Variables.
2. Under System variables, click New.
  - Variable name: JAVA\_HOME
  - Variable value: C:\Program Files\Eclipse Adoptium\jdk-17.0.18-hotspot (your Java folder path)
  - We then click Ok
3. Add Java to Path:
  - In the same Environment Variables window, select the variable Path under System variables → Edit → New.
  - We add this to the line: %JAVA\_HOME%\bin
  - We then click ok
4. To verify, we open the terminal and run java -version and echo %JAVA\_HOME%

```
(taxic) C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream>java --version
openjdk 17.0.18 2026-01-20
OpenJDK Runtime Environment Temurin-17.0.18+8 (build 17.0.18+8)
OpenJDK 64-Bit Server VM Temurin-17.0.18+8 (build 17.0.18+8, mixed mode, sharing)

(taxic) C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream>echo %JAVA_HOME%
C:\Program Files\Eclipse Adoptium\jdk-17.0.18-hotspot
```

### STEP VIII: SPARK STREAMING

1. We now run this command in the terminal:  
>python spark/stream\_processor.py
2. We also run this command, because even with Java installed, Spark still needs the Kafka connector.  
>spark-submit --packages org.apache.spark:spark-sql-kafka-0-10\_2.12:3.5.8  
spark/stream\_processor.py  
We will mostly be using this command.  
But, this will not work unless the following steps are completed too:
3. We go to the link <https://spark.apache.org/downloads.html>, and download Spark 3.5.8, Spark 3.5.8
4. A file gets downloaded: spark-3.5.8-bin-hadoop3
5. We Go to MY PC-> Local Disk C (C:\) ->Create a folder ‘spark’, and put the file ‘spark-3.5.8-bin-hadoop3’ inside it.
6. We now copy this file path
7. We go to Windows Search->Environment Variable, and under “System Variable”, we make:
  - Variable name: SPARK\_HOME
  - Value: C:\spark\spark-3.5.8-bin-hadoop3
  - Click Ok
8. Under System Variable itself, we find “Path”, we go there, and click Edit->New->and add: %SPARK\_HOME%\bin. Then: Ok Ok Ok.
9. In the terminal, we run command:  
>set PYSPARK\_PYTHON=C:\Users\rajve\anaconda3\envs\taxic\python.exe

-> This is needed because Spark doesn't automatically detect the Anaconda Python inside your `taxic` environment.

10. Now, in the Local C, we again create a folder: `hadoop`, and inside it, the folder: `bin`

11. We go to the link:

<https://github.com/cdarlint/winutils/tree/master/hadoop-3.2.2/bin>

12. From there we download both `winutils.exe` and `hadoop.dll`.

13. Now we Set Environment Variables: We go to System Variable->New, and then add:

- Variable name: `HADOOP_HOME`
- Value: `C:\hadoop`

14. In the System Variable itself, we find the "path" option, there we go to edit->new->and add: `C:\hadoop\bin`, or, `%HADOOP_HOME%\bin`

### STEP IX: TIME TO USE TWO TERMINALS TOGETHER

1. In Terminal 1, run:

```
>set PYSPARK_PYTHON=C:\Users\rajve\anaconda3\envs\taxic\python.exe
```

```
>spark-submit --packages org.apache.spark:spark-sql-kafka-0-10_2.12:3.5.8
```

`spark/stream_processor.py`

It will show as:

```

Batch: 0

+-----+-----+
|avg(fare_amount)|avg(trip_distance)|
+-----+-----+
| | |
+-----+-----+
```

But it won't start as of now.

2. In terminal 2:

```
>python producer/producer.py
```

As this goes on:

```
Sent ride 0
Sent ride 1
Sent ride 2
Sent ride 3
Sent ride 4
```

So does this too:

```
Batch: 1

+-----+-----+
|avg(fare_amount)|avg(trip_distance)|
+-----+-----+
| 10.0| 1.6|
+-----+-----+

Batch: 2

+-----+-----+
|avg(fare_amount)|avg(trip_distance)|
+-----+-----+
| 8.6| 1.06375|
+-----+-----+

Batch: 3

+-----+-----+
| avg(fare_amount)|avg(trip_distance)|
+-----+-----+
|10.053846153846154|1.3738461538461537|
+-----+-----+

Batch: 4

+-----+-----+
```

3. We can have a Quick Verification by going on the browser, and typing:

><http://localhost:4040>

-> This is the **Apache Spark UI**. There you can see:

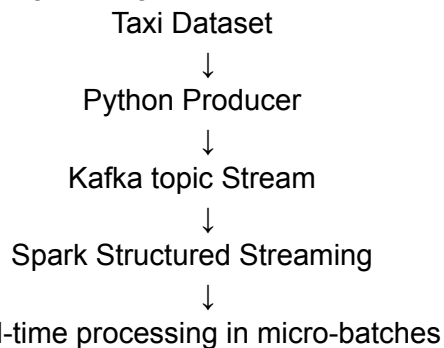
- Jobs
- Stages
- Streaming queries
- Execution DAG
- Processing times

->This is very useful when demonstrating the project.

| Job ID (Job Group)                        | Description                                                                                                                                 | Submitted           | Duration | Stages: Succeeded/Total | Tasks (for all stages): Succeeded/Total |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------|-------------------------|-----------------------------------------|
| 11 (e37bcdef-a18f-4ded-826c-46d24b874ac6) | id = add257f3-284b-4d73-9d5e-45cb3342c26 runId = e37bcdef-a18f-4ded-826c-46d24b874ac6 batch = 11<br>start at NativeMethodAccessonImpl.java0 | 2026/03/15 02:03:57 | 1 s      | 2/2                     | 2/2                                     |
| 10 (e37bcdef-a18f-4ded-826c-46d24b874ac6) | id = add257f3-284b-4d73-9d5e-45cb3342c26 runId = e37bcdef-a18f-4ded-826c-46d24b874ac6 batch = 10<br>start at NativeMethodAccessonImpl.java0 | 2026/03/15 02:03:55 | 0.9 s    | 2/2                     | 2/2                                     |
| 9 (e37bcdef-a18f-4ded-826c-46d24b874ac6)  | id = add257f3-284b-4d73-9d5e-45cb3342c26 runId = e37bcdef-a18f-4ded-826c-46d24b874ac6 batch = 9<br>start at NativeMethodAccessonImpl.java0  | 2026/03/15 02:03:52 | 1.0 s    | 2/2                     | 2/2                                     |
| 8 (e37bcdef-a18f-4ded-826c-46d24b874ac6)  | id = add257f3-284b-4d73-9d5e-45cb3342c26 runId = e37bcdef-a18f-4ded-826c-46d24b874ac6 batch = 8<br>start at NativeMethodAccessonImpl.java0  | 2026/03/15 02:03:49 | 1 s      | 2/2                     | 2/2                                     |
| 7 (e37bcdef-a18f-4ded-826c-46d24b874ac6)  | id = add257f3-284b-4d73-9d5e-45cb3342c26 runId = e37bcdef-a18f-4ded-826c-46d24b874ac6 batch = 7<br>start at NativeMethodAccessonImpl.java0  | 2026/03/15 02:03:47 | 1 s      | 2/2                     | 2/2                                     |
| 6 (e37bcdef-a18f-4ded-826c-46d24b874ac6)  | id = add257f3-284b-4d73-9d5e-45cb3342c26 runId = e37bcdef-a18f-4ded-826c-46d24b874ac6 batch = 6<br>start at NativeMethodAccessonImpl.java0  | 2026/03/15 02:03:44 | 1 s      | 2/2                     | 2/2                                     |
| 5 (e37bcdef-a18f-4ded-826c-46d24b874ac6)  | id = add257f3-284b-4d73-9d5e-45cb3342c26 runId = e37bcdef-a18f-4ded-826c-46d24b874ac6 batch = 5<br>start at NativeMethodAccessonImpl.java0  | 2026/03/15 02:03:42 | 1 s      | 2/2                     | 2/2                                     |
| 4 (e37bcdef-a18f-4ded-826c-46d24b874ac6)  | id = add257f3-284b-4d73-9d5e-45cb3342c26 runId = e37bcdef-a18f-4ded-826c-46d24b874ac6 batch = 4<br>start at NativeMethodAccessonImpl.java0  | 2026/03/15 02:03:39 | 0.7 s    | 2/2                     | 2/2                                     |
| 3 (e37bcdef-a18f-4ded-826c-46d24b874ac6)  | id = add257f3-284b-4d73-9d5e-45cb3342c26 runId = e37bcdef-a18f-4ded-826c-46d24b874ac6 batch = 3<br>start at NativeMethodAccessonImpl.java0  | 2026/03/15 02:03:37 | 1 s      | 2/2                     | 2/2                                     |
| 2 (e37bcdef-a18f-4ded-826c-46d24b874ac6)  | id = add257f3-284b-4d73-9d5e-45cb3342c26 runId = e37bcdef-a18f-4ded-826c-46d24b874ac6 batch = 2<br>start at NativeMethodAccessonImpl.java0  | 2026/03/15 02:03:35 | 1 s      | 2/2                     | 2/2                                     |
| 1 (e37bcdef-a18f-4ded-826c-46d24b874ac6)  | id = add257f3-284b-4d73-9d5e-45cb3342c26 runId = e37bcdef-a18f-4ded-826c-46d24b874ac6 batch = 1<br>start at NativeMethodAccessonImpl.java0  | 2026/03/15 02:03:31 | 2 s      | 2/2                     | 2/2                                     |
| 0 (e37bcdef-a18f-4ded-826c-46d24b874ac6)  | id = add257f3-284b-4d73-9d5e-45cb3342c26 runId = e37bcdef-a18f-4ded-826c-46d24b874ac6 batch = 0<br>start at NativeMethodAccessonImpl.java0  | 2026/03/15 02:02:57 | 3 s      | 1/1 (1 skipped)         | 1/1                                     |

## BRIEF OF WHAT WE HAVE DONE TILL NOW:

1. We now have a real-time data engineering pipeline:



Using:

- Apache Kafka
- Apache Spark
- Apache Spark Structured Streaming

2. Now, in short,

For Terminal 1:

```

>activate taxic
>cd "C:\Users\rjave\Desktop\SkillIssues\Personal Projects\taxistream"
>docker ps
>docker compose up -d
>set PYSPARK_PYTHON=C:\Users\rjave\anaconda3\envs\taxic\python.exe
>spark-submit --packages org.apache.spark:spark-sql-kafka-0-10_2.12:3.5.8

```

spark/stream\_processor.py

And for Terminal 2:

>activate taxic

>cd "C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream"

>python producer/producer.py

3. Completed stages:

1. Environment setup
2. Spark installation
3. Hadoop utilities setup
4. Kafka services started with Docker
5. Producer script sending taxi ride events
6. Spark streaming job consuming Kafka data
7. Micro-batch processing working

The **end-to-end pipeline is functional**.

4. Till here, this project demonstrates a modern real-time data engineering pipeline that;

- simulates live data streams
- transports data via Kafka
- processes data using Spark streaming
- operates in micro-batch streaming mode

->It showcases the fundamental architecture used in large-scale data systems such as ride-sharing platforms, financial trading systems, and IoT analytics.

5. Skills that have been demonstrated till yet in this project:

- Streaming Data Engineering
- Kafka Event Pipelines
- Spark Structured Streaming
- Dockerized Infrastructure
- Real-time Data Processing
- Data Pipeline Architecture

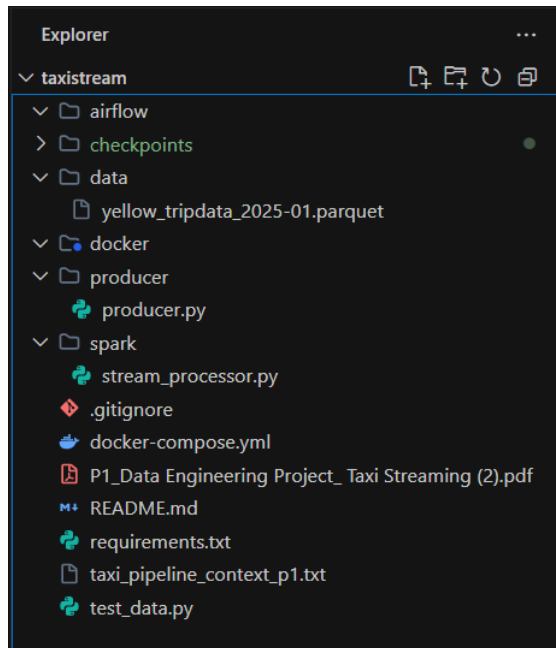
---

### **BRIEF OF WHAT HAS TO BE DONE NOW:**

1. Data storage in apache Cassandra
  2. Workflow orchestration in Apache Airflow for scheduling pipelines.
  3. Create analytics dashboards using Power BI or Grafana
  4. Continuous streaming: To Modify the producer so it continuously streams events instead of stopping at 50 records.
-

## STEP X: THE CASSANDRA BUILDING STAGE : PART 1 : UPDATING CODES

1. Before starting, this is how our project file looks like yet:



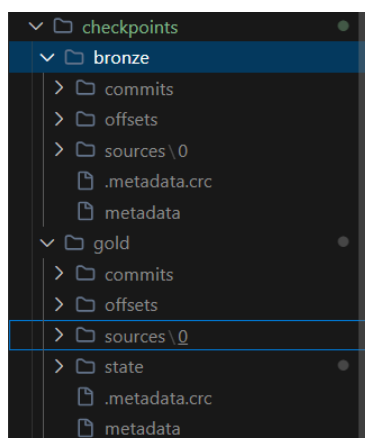
Where;

- i) The Airflow folder will be used for workflow orchestration. This folder will contain Apache Airflow DAGs. It will be used to: Schedule the pipeline, Start Spark jobs, Run batch tasks, Monitor pipelines. Right now it may be empty, but it is meant for automation.
- ii) The Checkpoints folder is for Spark Structured Streaming state storage. Spark uses this folder to store: Kafka offsets, Streaming progress, Metadata, State store data. Why does it exist? Without checkpoints, Spark cannot know: which Kafka messages were already processed, or, where to resume after restart. So this folder enables fault tolerance and recovery. If deleted, we need to follow these steps:

-> Restart Docker Desktop: Open task manager, and end these if they exist:

- Docker Desktop
- Docker Desktop Backend
- Com.docker.backend

Then we start the Docker Desktop again.



iii) The Data folder stores the NYC dataset: yellow\_tripdata\_2025-01.parquet

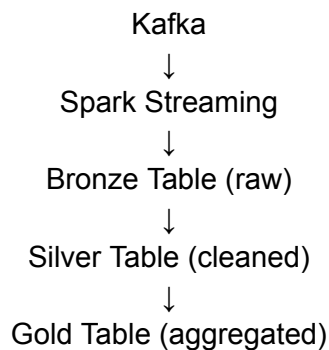
iv) The Producer folder is one of the main folders, and it is used for Kafka data ingestion. The producer.py file reads the taxi datasets, converts rows into JSON events, and sends events to kafka. So this script simulates real-time taxi ride events.

v) The Spark folder is one of the main folders used, and acts as the Streaming processing layer.

The file: stream\_processor.py is the main streaming engine. It:

- Reads events from Kafka
- Parses JSON
- Processes data
- Writes to Cassandra

-> Pipeline inside this script:



-> This implements the Medallion Architecture.

vi) The .gitignore file prevents unnecessary files from being uploaded such as:

- checkpoints/
- \_\_pycache\_\_/
- .env
- \*.log

vii) The docker-compose.yml file is one of the main files, which starts the whole infrastructure.

Till this stage of the project, it launches Kafka, Zookeeper, and Cassandra. The command: docker compose up -d creates the local data platform.

viii) The test\_data.py is basically used for testing / experimentation to inspect dataset, test transformations, and debug Kafka events. It isn't part of the main project.

2. These are the codes in every main file being used. Though only the stream\_processor.py file has been updated for the cassandra stage, just for the sake of documentation, here are the codes, as they should be for the working of this stage:

2.1) **producer.py :**

```
from kafka import KafkaProducer
import pandas as pd
import json
import time

Kafka connection
producer = KafkaProducer(
```

```

bootstrap_servers='localhost:9092',
value_serializer=lambda v: json.dumps(v).encode('utf-8')
)

Load dataset
df = pd.read_parquet('data/yellow_tripdata_2025-01.parquet')

Limit rows for testing
df = df.head(50)

topic = "taxi_rides"

for index, row in df.iterrows():

 data = row.to_dict()

 # Convert timestamps to string
 for key, value in data.items():
 if hasattr(value, "isoformat"):
 data[key] = value.isoformat()

 producer.send(topic, value=data)

 print(f"Sent ride {index}")

 time.sleep(0.5) # simulate real-time streaming

```

Only 49 rides are being sent for the time being

2.2) **docker-compose.yml** :

```

services:

 zookeeper:
 image: confluentinc/cp-zookeeper:7.5.0
 container_name: zookeeper
 environment:
 ZOOKEEPER_CLIENT_PORT: 2181
 ports:
 - "2181:2181"

 kafka:

```

```

 image: confluentinc/cp-kafka:7.5.0
 container_name: kafka
 depends_on:
 - zookeeper
 ports:
 - "9092:9092"
 environment:
 KAFKA_BROKER_ID: 1
 KAFKA_ZOOKEEPER_CONNECT: zookeeper:2181
 KAFKA_ADVERTISED_LISTENERS: PLAINTEXT://localhost:9092
 KAFKA_OFFSETS_TOPIC_REPLICATION_FACTOR: 1
 cassandra:
 image: cassandra:4.1
 container_name: cassandra
 ports:
 - "9042:9042"
 environment:
 - CASSANDRA_CLUSTER_NAME=TaxiCluster
 volumes:
 - cassandra_data:/var/lib/cassandra

volumes:
 cassandra_data:

```

**2.3) stream\_processor.py** : This is where real change has been made:

```

from pyspark.sql import SparkSession
from pyspark.sql.functions import to_date
from pyspark.sql.functions import col, from_json, current_timestamp, expr,
avg, count, to_timestamp
from pyspark.sql.types import StructType, StringType, DoubleType,
IntegerType
from pyspark.sql.functions import window

----- Spark Session -----
spark = SparkSession.builder \
 .appName("TaxiStreamingProcessor") \
 .config("spark.sql.shuffle.partitions", "2") \
 .config("spark.cassandra.connection.host", "127.0.0.1") \
 .config("spark.driver.host", "127.0.0.1") \
 .config("spark.sql.streaming.stateStore.providerClass",

```

```

"org.apache.spark.sql.execution.streaming.state.HDFSBackedStateStoreProvider") \
 .getOrCreate()

spark.catalog.clearCache()
spark.sparkContext.setLogLevel("WARN")

----- Full schema (all fields from your Kafka JSON)

schema = StructType() \
 .add("VendorID", IntegerType()) \
 .add("tpep_pickup_datetime", StringType()) \
 .add("tpep_dropoff_datetime", StringType()) \
 .add("passenger_count", DoubleType()) \
 .add("trip_distance", DoubleType()) \
 .add("RatecodeID", DoubleType()) \
 .add("store_and_fwd_flag", StringType()) \
 .add("PULocationID", IntegerType()) \
 .add("DOLocationID", IntegerType()) \
 .add("payment_type", IntegerType()) \
 .add("fare_amount", DoubleType()) \
 .add("extra", DoubleType()) \
 .add("mta_tax", DoubleType()) \
 .add("tip_amount", DoubleType()) \
 .add("tolls_amount", DoubleType()) \
 .add("improvement_surcharge", DoubleType()) \
 .add("total_amount", DoubleType()) \
 .add("congestion_surcharge", DoubleType()) \
 .add("Airport_fee", DoubleType()) \
 .add("cbd_congestion_fee", DoubleType())

----- Read Kafka Stream -----
df = spark.readStream \
 .format("kafka") \
 .option("kafka.bootstrap.servers", "localhost:9092") \
 .option("subscribe", "taxi_rides") \
 .option("startingOffsets", "earliest") \
 .load()

```

```

json_df = df.selectExpr("CAST(value AS STRING) AS raw_event") \
 .withColumn("ride_id", expr("uuid()"))
----- BRONZE -----
bronze_df = json_df.withColumn("created_at", current_timestamp())
bronze_query = bronze_df.writeStream \
 .foreachBatch(lambda batch_df, _: batch_df.write
 .format("org.apache.spark.sql.cassandra")
 .options(table="bronze_rides", keyspace="taxi_streaming")
 .mode("append")
 .save()
) \
 .option("checkpointLocation", "checkpoints/bronze") \
 .outputMode("append") \
 .start()

----- SILVER -----
silver_df = json_df.select(
 from_json(col("raw_event"), schema).alias("data"),
 col("ride_id")
).select("ride_id", "data.*")

silver_df = silver_df.select(
 col("ride_id"),
 col("VendorID").alias("vendorid"),
 col("tpep_pickup_datetime"),
 col("tpep_dropoff_datetime"),
 col("passenger_count"),
 col("trip_distance"),
 col("RatecodeID").alias("ratecodeid"),
 col("store_and_fwd_flag"),
 col("PULocationID").alias("pulocationid"),
 col("DOLocationID").alias("dolocationid"),
 col("payment_type"),
 col("fare_amount"),
 col("extra"),
 col("mta_tax"),
 col("tip_amount"),
 col("tolls_amount"),
 col("improvement_surcharge"),
 col("total_amount"),

```



|                  |              |                 |      |
|------------------|--------------|-----------------|------|
| ca-certificates  | 2025.12.2    | haa95532_0      |      |
| cassandra-driver | 3.29.3       | pypi_0          | pypi |
| click            | 8.3.1        | pypi_0          | pypi |
| colorama         | 0.4.6        | pypi_0          | pypi |
| expat            | 2.7.4        | hd7fb8db_0      |      |
| findspark        | 2.0.1        | pypi_0          | pypi |
| geomet           | 1.1.0        | pypi_0          | pypi |
| kafka-python     | 2.3.0        | pypi_0          | pypi |
| libexpat         | 2.7.4        | hd7fb8db_0      |      |
| libffi           | 3.4.4        | hd77b12b_1      |      |
| libzlib          | 1.3.1        | h02ab6af_0      |      |
| numpy            | 2.2.6        | pypi_0          | pypi |
| openssl          | 3.5.5        | hbb43b14_0      |      |
| packaging        | 25.0         | py310haa95532_1 |      |
| pandas           | 2.3.3        | pypi_0          | pypi |
| pip              | 26.0.1       | pyhc872135_0    |      |
| py4j             | 0.10.9.9     | pypi_0          | pypi |
| pyarrow          | 23.0.1       | pypi_0          | pypi |
| pyspark          | 4.1.1        | pypi_0          | pypi |
| python           | 3.10.19      | h981015d_0      |      |
| python-dateutil  | 2.9.0.post0  | pypi_0          | pypi |
| pytz             | 2026.1.post1 | pypi_0          | pypi |
| setuptools       | 80.10.2      | py310haa95532_0 |      |
| six              | 1.17.0       | pypi_0          | pypi |
| sqlite           | 3.51.1       | hee5a0db_1      |      |
| tk               | 8.6.15       | hf199647_0      |      |
| tzdata           | 2025.3       | pypi_0          | pypi |
| ucrt             | 10.0.22621.0 | haa95532_0      |      |
| vc               | 14.3         | h2df5915_10     |      |
| vc14_runtime     | 14.44.35208  | h4927774_10     |      |
| vs2015_runtime   | 14.44.35208  | ha6b5a95_10     |      |
| wheel            | 0.46.3       | py310haa95532_0 |      |
| xz               | 5.8.2        | h53af0af_0      |      |
| zlib             | 1.3.1        | h02ab6af_0      |      |

## STEP XI: THE CASSANDRA BUILDING STAGE : PART 2: USING CASSANDRA SHELL

We follow the required steps to create tables for bronze, silver, and gold levels. (Bronze and Gold are stored in Cassandra, Silver exists only in Spark):

1. We start docker: kafka, zookeeper, cassandra, in the terminal:

```
> docker compose up -d
```

2. We open the cassandra shell:

```
> docker exec -it cassandra cqlsh
```

3. We create a database for the tables in this project:

```
CREATE KEYSPACE taxi_streaming WITH replication = {'class': 'SimpleStrategy',
'replication_factor': '1'} AND durable_writes = true;
```

4. We check the whether the database has been made or not:

```
cqlsh> DESCRIBE KEYSPACES;
```

```
cqlsh> DESCRIBE KEYSPACES;
```

```
my_keyspace system_distributed system_views taxi_streaming
system system_schema system_virtual_schema
system_auth system_traces taf
```

5. We use this database:

```
cqlsh>USE taxi_streaming;
```

```
cqlsh:taxi_streaming>
```

6. We create the tables inside it:

6.1) **For bronze layer:**

```
cqlsh> CREATE TABLE bronze_rides (
 ride_id text PRIMARY KEY,
 created_at timestamp,
 raw_event text
);
```

6.2) **For silver layer:**

```
cqlsh>CREATE TABLE silver_rides (
 ride_id text PRIMARY KEY,
 airport_fee double,
 cbd_congestion_fee double,
 congestion_surcharge double,
 dolocationid int,
 dropoff_datetime timestamp,
 extra double,
 fare_amount double,
 improvement_surcharge double,
 mta_tax double,
 passenger_count double,
 payment_type int,
 pickup_datetime timestamp,
 pulocationid int,
```

```

ratecodeid double,
store_and_fwd_flag text,
tip_amount double,
tolls_amount double,
total_amount double,
tpep_dropoff_datetime text,
tpep_pickup_datetime text,
trip_distance double,
vendorid int
);

```

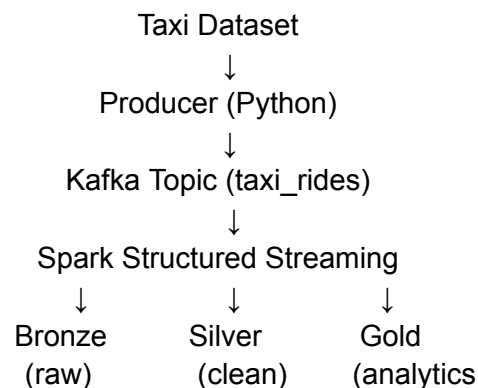
### 6.3) For Gold Layer:

```

cqlsh>CREATE TABLE gold_rides (
 agg_date date,
 agg_time timestamp,
 avg_fare double,
 avg_trip_distance double,
 ride_count bigint,
 PRIMARY KEY (agg_date, agg_time)
) WITH CLUSTERING ORDER BY (agg_time DESC);

```

- > The Bronze Layer is used to Store raw Kafka messages, involves no cleaning, and is basically just ingestion of data
- > The Silver Layer parses JSON schema, and cleans and structures taxi ride fields
- > The Gold Layer performs streaming aggregation analytics, and computes queries involving avg fare, avg trip distance, or, ride count per time window.
- > The Gold table is partitioned by agg\_date and clustered by agg\_time so that analytics queries can efficiently retrieve recent ride metrics by day.
- > The Gold layer also uses windowing for real-time analysis and time-based aggregation (per minute)
- > Hence the architecture represents the **Medallion Architecture**, used in modern data engineering pipelines.



#### NOTE:

Bronze and Gold layers are persisted in Cassandra.  
Silver layer exists only inside Spark for transformations.

6.4) To check the tables and its structure, we run the command:

cqlsh>DESCRIBE taxi\_streaming;

```
(taxic) C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream>docker exec -it cassandra cqlsh
Connected to TaxiCluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.10 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh> DESCRIBE taxi_streaming;

CREATE KEYSPACE taxi_streaming WITH replication = {'class': 'SimpleStrategy', 'replication_factor': '1'} AND durable_writes = true;

CREATE TABLE taxi_streaming.bronze_rides (
 ride_id text PRIMARY KEY,
 created_at timestamp,
 raw_event text
) WITH additional_write_policy = '99p'
 AND bloom_filter_fp_chance = 0.01
 AND caching = {'keys': 'ALL', 'rows_per_partition': 'NONE'}
 AND cdc = false
 AND comment = ''
 AND compaction = {'class': 'org.apache.cassandra.db.compaction.SizeTieredCompactionStrategy', 'max_threshold': '32', 'min_threshold': '4'}
 AND compression = {'chunk_length_in_kb': '16', 'class': 'org.apache.cassandra.io.compress.LZ4Compressor'}
 AND memtable = 'default'
 AND crc_check_chance = 1.0
 AND default_time_to_live = 0
 AND extensions = {}
 AND gc_grace_seconds = 864000
 AND max_index_interval = 2048
 AND memtable_flush_period_in_ms = 0
 AND min_index_interval = 128
 AND read_repair = 'BLOCKING'
 AND speculative_retry = '99p';

CREATE TABLE taxi_streaming.gold_rides (
 agg_date date,
 agg_time timestamp,
 avg_fare double,
 avg_trip_distance double,
 ride_count bigint,
 PRIMARY KEY (agg_date, agg_time)
) WITH CLUSTERING ORDER BY (agg_time DESC)
 AND additional_write_policy = '99p'
 AND bloom_filter_fp_chance = 0.01
 AND caching = {'keys': 'ALL', 'rows_per_partition': 'NONE'}
 AND cdc = false
 AND comment = ''
 AND compaction = {'class': 'org.apache.cassandra.db.compaction.SizeTieredCompactionStrategy', 'max_threshold': '32', 'min_threshold': '4'}
 AND compression = {'chunk_length_in_kb': '16', 'class': 'org.apache.cassandra.io.compress.LZ4Compressor'}
 AND memtable = 'default'
 AND crc_check_chance = 1.0
 AND default_time_to_live = 0
 AND extensions = {}
 AND gc_grace_seconds = 864000
 AND max_index_interval = 2048
 AND memtable_flush_period_in_ms = 0
 AND min_index_interval = 128
 AND read_repair = 'BLOCKING'
 AND speculative_retry = '99p';
```

```
CREATE TABLE taxi_streaming.silver_rides (
 ride_id text PRIMARY KEY,
 airport_fee double,
 cbd_congestion_fee double,
 congestion_surcharge double,
 dolocationid int,
 dropoff_datetime timestamp,
 extra double,
 fare_amount double,
 improvement_surcharge double,
 mta_tax double,
 passenger_count double,
 payment_type int,
 pickup_datetime timestamp,
 pulocationid int,
 ratecodeid double,
 store_and_fwd_flag text,
 tip_amount double,
 tolls_amount double,
 total_amount double,
 tpep_dropoff_datetime text,
 tpep_pickup_datetime text,
 trip_distance double,
 vendorid int
) WITH additional_write_policy = '99p'
 AND bloom_filter_fp_chance = 0.01
 AND caching = {'keys': 'ALL', 'rows_per_partition': 'NONE'}
 AND cdc = false
 AND comment = ''
 AND compaction = {'class': 'org.apache.cassandra.db.compaction.SizeTieredCompactionStrategy', 'max_threshold': '32', 'min_threshold': '4'}
 AND compression = {'chunk_length_in_kb': '16', 'class': 'org.apache.cassandra.io.compress.LZ4Compressor'}
 AND memtable = 'default'
 AND crc_check_chance = 1.0
 AND default_time_to_live = 0
 AND extensions = {}
 AND gc_grace_seconds = 864000
 AND max_index_interval = 2048
 AND memtable_flush_period_in_ms = 0
 AND min_index_interval = 128
 AND read_repair = 'BLOCKING'
 AND speculative_retry = '99p';
cqlsh>
```

NOTE: We ensure that the columns made here in the cqlsh database, and the ones written in the stream\_processor.py file are the same. Otherwise Spark may try writing into columns, which Cassandra might not have.

## **STEP XII: USING CASSANDRA**

We assume that Docker Desktop is down, and no 'checkpoints' file is made yet. This is actually how this stage of the project started. But anyhow just for demonstration, we may delete the checkpoints folder, and also turn off the docker engine, to show how to start the pipeline from the very beginning (because we can't always keep our docker on)

Also, remember:

1. If we delete the Checkpoint folder, we again have to restart Docker, and kafka containers.
2. If we had turned off the docker down, we would have to delete the checkpoint folder.

This happens because Spark stores Kafka offsets inside the checkpoint folder. Example:

Checkpoint says: 150

But if we run docker compose down, Kafka restarts and the topic starts again at:offset: 0. Now spark sees: expected offset: 150, and actual offset: 0. Spark thinks data was lost, so it throws the offset error we saw.

Anyway, we start with how to proceed with the project, when no checkpoint folder is there yet, or if we delete the checkpoint folder:

### **CASE I : WHEN THERE IS NO CHECKPOINT FOLDER YET:**

1. We delete the checkpoint folder from the main project file taxistream, if it's already present.
2. We go to Task Manager, and end these tasks if they exist:
  - Docker Desktop
  - Docker Desktop Backend
  - com.docker.backend

Then we start Docker Desktop again, just by clicking on the Docker Desktop icon.

3. Now, in Terminal 1, we run the follow commands:

```
>activate taxic
```

```
>cd "C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream"
```

```
>docker compose down
```

```
>docker compose up -d
```

```
>docker exec -it kafka kafka-topics --create --topic taxi_rides --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1
```

(to create a kafka topic, has to be done everytime the docker has been up, after down )

```
>docker exec -it kafka kafka-topics --list --bootstrap-server localhost:9092
```

(to verify whether docker is running or not)

```
(base) C:\Users\rajve>activate taxic
(taxic) C:\Users\rajve>cd "C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream"
(taxic) C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream>docker compose down
[+] down 4/4
✓Container cassandra Removed 0.1s
✓Container kafka Removed 0.1s
✓Container zookeeper Removed 0.1s
✓Network taxistream_default Removed 0.3s
(taxic) C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream>docker compose up -d
[+] up 4/4
✓Network taxistream_default Created 0.1s
✓Container cassandra Started 1.3s
✓Container zookeeper Started 1.3s
✓Container kafka Started 1.4s
(taxic) C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream>docker exec -it kafka kafka-topics --list --bootstrap-server localhost:9092
(taxic) C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream>docker exec -it kafka kafka-topics --create --topic taxi_rides --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid issues it is best to use either, but not both.
Created topic taxi_rides.
```

>set PYSPARK\_PYTHON=C:\Users\rajve\anaconda3\envs\taxic\python.exe

>spark-submit --packages

org.apache.spark:spark-sql-kafka-0-10\_2.12:3.5.8,com.datastax.spark:spark-cassandra-connector\_2.12:3.5.0 spark/stream\_processor.py

And then basically it'll go up to somewhere like:

```
26/03/16 23:19:58 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, 127.0.0.1, 62535, None)
26/03/16 23:19:58 INFO SharedState: Setting hive.metastore.warehouse.dir ('null') to the value of spark.sql.warehouse.dir.
26/03/16 23:19:58 INFO SharedState: Warehouse path is 'file:/C:/Users/rajve/Desktop/SkillIssues/Personal%20Projects/taxistream/spark-warehouse/'.
26/03/16 23:20:03 WARN ResolveWriteToStream: spark.sql.adaptive.enabled is not supported in streaming DataFrames/Datasets and will be disabled.
26/03/16 23:20:04 WARN AdminClientConfig: These configurations '[key.deserializer, value.deserializer, enable.auto.commit, max.poll.records, auto.offset.reset]' were supplied but are not used yet.
26/03/16 23:20:04 WARN ResolveWriteToStream: spark.sql.adaptive.enabled is not supported in streaming DataFrames/Datasets and will be disabled.
26/03/16 23:20:04 WARN AdminClientConfig: These configurations '[key.deserializer, value.deserializer, enable.auto.commit, max.poll.records, auto.offset.reset]' were supplied but are not used yet.
```

.....

4. Meanwhile in Terminal 2:

>activate taxic

>cd "C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream"

>python producer/producer.py

It'll send rides:

```
(taxic) C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream>python producer/producer.py
Sent ride 0
Sent ride 1
Sent ride 2
Sent ride 3
Sent ride 4
Sent ride 5
Sent ride 6
Sent ride 7
Sent ride 8
```

.....

To Verify, We go to local host:

<http://localhost:4040/jobs/> ( [TaxiStreamingProcessor - Spark Jobs](#) )

Here, we'll be able to see the Spark UI:

[illegible]

For the first time, 20,22,26 jobs are expected, for 49 rides being sent.

We can also check whether Cassandra is giving results or not, by going to the `cqlsh` shell. This is also to be done in the terminal 2, as terminal 1 is still running `spark`.

```
>docker exec -t cassandra cqlsh
```

```
cqlsh>USE taxi_streaming;
```

```
cqlsh> SELECT * FROM taxi_streaming.bronze_rides LIMIT 5;
```

```
cqlsh> SELECT * FROM taxi_streaming.silver_rides LIMIT 5;
```

```
cqlsh>SELECT * FROM taxi_streaming.gold_rides WHERE agg_date = '2026-03-16'
```

LIMIT 10;

And notice that calling the bronze table only gives raw data, calling the silver layer gives the cleaned structure, and the gold layer gives proper analysis.

```

cqlsh> SELECT * FROM taxi_streaming.bronze_rides LIMIT 5;

ride_id	created_at	raw_event
1d958d63-dd9a-46b7-b7ad-ddd142d9c91b | 2026-03-16 11:22:06.198000+0000 | {"VendorID": 2, "tpep_pickup_datetime": "2025-01-01T00:14:27", "tpep_dropoff_datetime": "2025-01-01T00:20:01", "passenger_count": 3.0, "trip_distance": 0.52, "RatecodeID": 1.0, "store_and_fwd_flag": "N", "PULocationID": 244, "DOLocationID": 244, "payment_type": 2, "fare_amount": 7.2, "extra": 1.0, "mta_tax": 0.5, "tip_amount": 0.0, "tolls_amount": 0.0, "improvement_surcharge": 1.0, "total_amount": 9.7, "congestion_surcharge": 0.0, "Airport_fee": 0.0, "cbd_congestion_fee": 0.0}
675d7388-e97b-4d5c-ba0c-df2fb0ff7e1b0 | 2026-03-16 17:45:57.893000+0000 | {"VendorID": 2, "tpep_pickup_datetime": "2025-01-01T00:21:34", "tpep_dropoff_datetime": "2025-01-01T00:25:06", "passenger_count": 3.0, "trip_distance": 0.66, "RatecodeID": 1.0, "store_and_fwd_flag": "N", "PULocationID": 244, "DOLocationID": 116, "payment_type": 2, "fare_amount": 5.8, "extra": 1.0, "mta_tax": 0.5, "tip_amount": 0.0, "tolls_amount": 0.0, "improvement_surcharge": 1.0, "total_amount": 8.3, "congestion_surcharge": 0.0, "Airport_fee": 0.0, "cbd_congestion_fee": 0.0}
4244e035-d091-dd8b-bba1-21187b688946 | 2026-03-16 17:50:52.180000+0000 | {"VendorID": 1, "tpep_pickup_datetime": "2025-01-01T00:14:47", "tpep_dropoff_datetime": "2025-01-01T00:16:15", "passenger_count": 0.0, "trip_distance": 0.4, "RatecodeID": 1.0, "store_and_fwd_flag": "N", "PULocationID": 170, "DOLocationID": 170, "payment_type": 1, "fare_amount": 4.4, "extra": 2.5, "mta_tax": 0.5, "tip_amount": 2.35, "tolls_amount": 0.0, "improvement_surcharge": 1.0, "total_amount": 11.75, "congestion_surcharge": 2.5, "Airport_fee": 0.0, "cbd_congestion_fee": 0.0}
c02d7b46-8330-e47b-9457-8e1df9758830 | 2026-03-16 11:15:14.102000+0000 | {"VendorID": 2, "tpep_pickup_datetime": "2025-01-01T00:53:47", "tpep_dropoff_datetime": "2025-01-01T00:58:02", "passenger_count": 1.0, "trip_distance": 0.65, "RatecodeID": 1.0, "store_and_fwd_flag": "N", "PULocationID": 263, "DOLocationID": 141, "payment_type": 1, "fare_amount": 5.8, "extra": 1.0, "mta_tax": 0.5, "tip_amount": 2.16, "tolls_amount": 0.0, "improvement_surcharge": 1.0, "total_amount": 12.96, "congestion_surcharge": 2.5, "Airport_fee": 0.0, "cbd_congestion_fee": 0.0}
0ef31676-65e9-42ae-871e-eale24c61cda | 2026-03-16 12:53:27.496000+0000 | {"VendorID": 2, "tpep_pickup_datetime": "2025-01-01T00:01:41", "tpep_dropoff_datetime": "2025-01-01T00:07:14", "passenger_count": 1.0, "trip_distance": 0.71, "RatecodeID": 1.0, "store_and_fwd_flag": "N", "PULocationID": 79, "DOLocationID": 107, "payment_type": 2, "fare_amount": 7.2, "extra": 1.0, "mta_tax": 0.5, "tip_amount": 0.0, "tolls_amount": 0.0, "improvement_surcharge": 1.0, "total amount": 12.2, "congestion_surcharge": 2.5, "Airport_fee": 0.0, "cbd_congestion_fee": 0.0}

(5 rows)

```

```
cqlsh> SELECT * FROM taxi_streaming.silver_rides LIMIT 5;
```

| ride_id | airport_fee | cbd_congestion_fee | congestion_surcharge | dolocationid | dropoff_datetime    | extra | fare_amount | improvement_surcharge | mta_tax | passenger_count | payment_type | pickup_datetime     | pulocationid | ratecodeid | store_and_fwd_flag | tip_amount | tolls_amount | total_amount | tpep_dropoff_datetime | tpep_pickup_datetime | trip_distance | vendorid |
|---------|-------------|--------------------|----------------------|--------------|---------------------|-------|-------------|-----------------------|---------|-----------------|--------------|---------------------|--------------|------------|--------------------|------------|--------------|--------------|-----------------------|----------------------|---------------|----------|
| 1       | 0           | 0                  | 0                    | 1            | 2015-01-04 15:00:00 | 0     | 1.5         | 0                     | 0       | 1               | cash         | 2015-01-04 15:00:00 | 1            | 1          | 0                  | 0          | 0            | 1.5          | 2015-01-04 15:00:00   | 2015-01-04 15:00:00  | 0             | 1        |
| 2       | 0           | 0                  | 0                    | 1            | 2015-01-04 15:00:00 | 0     | 1.5         | 0                     | 0       | 1               | cash         | 2015-01-04 15:00:00 | 1            | 1          | 0                  | 0          | 0            | 1.5          | 2015-01-04 15:00:00   | 2015-01-04 15:00:00  | 0             | 1        |
| 3       | 0           | 0                  | 0                    | 1            | 2015-01-04 15:00:00 | 0     | 1.5         | 0                     | 0       | 1               | cash         | 2015-01-04 15:00:00 | 1            | 1          | 0                  | 0          | 0            | 1.5          | 2015-01-04 15:00:00   | 2015-01-04 15:00:00  | 0             | 1        |
| 4       | 0           | 0                  | 0                    | 1            | 2015-01-04 15:00:00 | 0     | 1.5         | 0                     | 0       | 1               | cash         | 2015-01-04 15:00:00 | 1            | 1          | 0                  | 0          | 0            | 1.5          | 2015-01-04 15:00:00   | 2015-01-04 15:00:00  | 0             | 1        |
| 5       | 0           | 0                  | 0                    | 1            | 2015-01-04 15:00:00 | 0     | 1.5         | 0                     | 0       | 1               | cash         | 2015-01-04 15:00:00 | 1            | 1          | 0                  | 0          | 0            | 1.5          | 2015-01-04 15:00:00   | 2015-01-04 15:00:00  | 0             | 1        |

```
cqlsh> SELECT * FROM taxi_streaming.gold_rides LIMIT 5;
```

| agg_date   | agg_time                        | avg_fare | avg_trip_distance | ride_count |
|------------|---------------------------------|----------|-------------------|------------|
| 2026-03-17 | 2026-03-16 20:36:55.983000+0000 | 9.3      | 1.63              | 1          |
| 2026-03-17 | 2026-03-16 20:36:52.852000+0000 | 27.5     | 4.26              | 1          |
| 2026-03-17 | 2026-03-16 20:36:49.998000+0000 | 12.45    | 1.725             | 3          |
| 2026-03-17 | 2026-03-16 20:36:47.372000+0000 | 13.15    | 1.665             | 2          |
| 2026-03-17 | 2026-03-16 20:36:44.428000+0000 | 0        | 0.69              | 2          |

(5 rows)

Some more examples:

```
cqlsh> SELECT * FROM taxi_streaming.gold_rides WHERE agg_date = '2026-03-16' LIMIT 10;
```

| agg_date   | agg_time                        | avg_fare | avg_trip_distance | ride_count |
|------------|---------------------------------|----------|-------------------|------------|
| 2026-03-16 | 2026-03-16 17:51:09.388000+0000 | 9.3      | 1.63              | 3          |
| 2026-03-16 | 2026-03-16 17:51:06.443000+0000 | 27.5     | 4.26              | 3          |
| 2026-03-16 | 2026-03-16 17:51:02.922000+0000 | 12.45    | 1.725             | 9          |
| 2026-03-16 | 2026-03-16 17:50:56.475000+0000 | 1.275    | 0.6175            | 8          |
| 2026-03-16 | 2026-03-16 17:50:53.470000+0000 | 13.78    | 1.94              | 5          |
| 2026-03-16 | 2026-03-16 17:46:16.832000+0000 | 9.3      | 1.63              | 2          |
| 2026-03-16 | 2026-03-16 17:46:13.225000+0000 | 7.9      | 0.7               | 2          |
| 2026-03-16 | 2026-03-16 17:46:10.076000+0000 | 12.45    | 1.725             | 6          |
| 2026-03-16 | 2026-03-16 17:46:03.043000+0000 | 1.02     | 0.632             | 5          |
| 2026-03-16 | 2026-03-16 17:45:59.479000+0000 | 14.66667 | 2.08333           | 3          |

```
cqlsh> SELECT * FROM taxi_streaming.gold_rides;
```

| agg_date   | agg_time                        | avg_fare | avg_trip_distance | ride_count |
|------------|---------------------------------|----------|-------------------|------------|
| 2026-03-17 | 2026-03-16 20:36:55.983000+0000 | 9.3      | 1.63              | 1          |
| 2026-03-17 | 2026-03-16 20:36:52.852000+0000 | 27.5     | 4.26              | 1          |
| 2026-03-17 | 2026-03-16 20:36:49.998000+0000 | 12.45    | 1.725             | 3          |
| 2026-03-17 | 2026-03-16 20:36:47.372000+0000 | 13.15    | 1.665             | 2          |
| 2026-03-17 | 2026-03-16 20:36:44.428000+0000 | 0        | 0.69              | 2          |
| 2026-03-17 | 2026-03-16 20:36:41.358000+0000 | 16.3     | 1.9               | 1          |
| 2026-03-17 | 2026-03-16 20:36:38.445000+0000 | 14.2     | 2.29              | 1          |
| 2026-03-17 | 2026-03-16 20:36:34.651000+0000 | 5.8      | 0.6               | 2          |
| 2026-03-17 | 2026-03-16 20:36:31.222000+0000 | 10       | 1.6               | 1          |
| 2026-03-16 | 2026-03-16 17:51:09.388000+0000 | 9.3      | 1.63              | 3          |
| 2026-03-16 | 2026-03-16 17:51:06.443000+0000 | 27.5     | 4.26              | 3          |
| 2026-03-16 | 2026-03-16 17:51:02.922000+0000 | 12.45    | 1.725             | 9          |
| 2026-03-16 | 2026-03-16 17:50:56.475000+0000 | 1.275    | 0.6175            | 8          |
| 2026-03-16 | 2026-03-16 17:50:53.470000+0000 | 13.78    | 1.94              | 5          |
| 2026-03-16 | 2026-03-16 17:46:16.832000+0000 | 9.3      | 1.63              | 2          |
| 2026-03-16 | 2026-03-16 17:46:13.225000+0000 | 7.9      | 0.7               | 2          |
| 2026-03-16 | 2026-03-16 17:46:10.076000+0000 | 12.45    | 1.725             | 6          |
| 2026-03-16 | 2026-03-16 17:46:03.043000+0000 | 1.02     | 0.632             | 5          |
| 2026-03-16 | 2026-03-16 17:45:59.479000+0000 | 14.66667 | 2.08333           | 3          |
| 2026-03-16 | 2026-03-16 17:40:02.796000+0000 | 9.3      | 1.63              | 1          |
| 2026-03-16 | 2026-03-16 17:39:58.854000+0000 | 27.5     | 4.26              | 1          |
| 2026-03-16 | 2026-03-16 17:39:54.626000+0000 | 1.7      | 0.593333          | 3          |
| 2026-03-16 | 2026-03-16 17:39:50.785000+0000 | 13.15    | 1.665             | 2          |
| 2026-03-16 | 2026-03-16 17:39:45.283000+0000 | 19.1     | 2.8               | 2          |
| 2026-03-16 | 2026-03-16 17:39:40.819000+0000 | 5.8      | 0.6               | 2          |
| 2026-03-16 | 2026-03-16 17:39:37.052000+0000 | 10       | 1.6               | 1          |
| 2026-03-16 | 2026-03-16 17:07:52.969000+0000 | 9.3      | 1.63              | 1          |
| 2026-03-16 | 2026-03-16 17:07:49.019000+0000 | 27.5     | 4.26              | 1          |
| 2026-03-16 | 2026-03-16 17:07:45.181000+0000 | 12.45    | 1.725             | 3          |
| 2026-03-16 | 2026-03-16 17:07:41.708000+0000 | 13.15    | 1.665             | 2          |
| 2026-03-16 | 2026-03-16 17:07:37.417000+0000 | 16.3     | 1.9               | 2          |
| 2026-03-16 | 2026-03-16 17:07:33.219000+0000 | 19.1     | 2.4               | 2          |
| 2026-03-16 | 2026-03-16 17:07:29.547000+0000 | 5.8      | 0.6               | 2          |
| 2026-03-16 | 2026-03-16 17:07:25.199000+0000 | 10       | 1.6               | 1          |
| 2026-03-16 | 2026-03-16 12:59:10.103000+0000 | 27.5     | 4.26              | 6          |
| 2026-03-16 | 2026-03-16 12:53:31.258000+0000 | 12.45    | 1.725             | 15         |
| 2026-03-16 | 2026-03-16 12:44:35.602000+0000 | 27.5     | 4.26              | 4          |
| 2026-03-16 | 2026-03-16 11:22:21.387000+0000 | 9.3      | 1.63              | 3          |
| 2026-03-16 | 2026-03-16 11:22:18.882000+0000 | 7.9      | 0.7               | 3          |
| 2026-03-16 | 2026-03-16 11:22:16.363000+0000 | 27.5     | 4.26              | 3          |
| 2026-03-16 | 2026-03-16 11:22:13.737000+0000 | 1.7      | 0.593333          | 9          |
| 2026-03-16 | 2026-03-16 11:22:08.831000+0000 | 13.78    | 1.94              | 8          |
| 2026-03-16 | 2026-03-16 11:15:20.592000+0000 | 9.3      | 1.63              | 2          |
| 2026-03-16 | 2026-03-16 11:15:17.863000+0000 | 7.9      | 0.7               | 2          |
| 2026-03-16 | 2026-03-16 11:15:14.966000+0000 | 27.5     | 4.26              | 2          |
| 2026-03-16 | 2026-03-16 11:15:11.727000+0000 | 1.7      | 0.593333          | 6          |
| 2026-03-16 | 2026-03-16 11:15:08.163000+0000 | 1.02     | 0.632             | 5          |
| 2026-03-16 | 2026-03-16 11:14:59.404000+0000 | 14.66667 | 2.08333           | 3          |
| 2026-03-16 | 2026-03-16 11:07:55.842000+0000 | 27.5     | 4.26              | 4          |

(49 rows)

## CASE II : WHEN THE CHECKPOINT FOLDER IS ALREADY THERE, AND DOCKER WAS NEVER DONE DOWN:

Suppose after the above process, we never did the docker down, basically it just remained up the whole time. In that case, since the docker was up the whole time, the checkpoints don't need to be deleted, and the kafka containers made are still active, even if the terminals are closed.

This will only work if docker wasn't done down and the checkpoints were not deleted. If any one was even done, we have to start from the beginning, that is, start docker, make new kafka containers, and run stream\_processor.py which will create new checkpoints.

Anyway, for Case II:

In terminal 1:

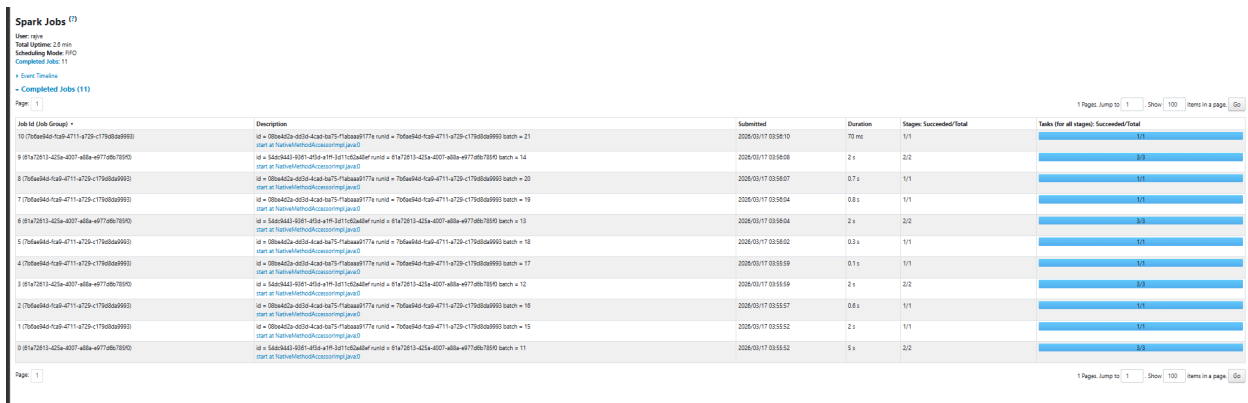
```
>activate taxic
>cd "C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream"
>set PYSARK_PYTHON=C:\Users\rajve\anaconda3\envs\taxic\python.exe
>spark-submit --packages
```

org.apache.spark:spark-sql-kafka-0-10\_2.12:3.5.8,com.datastax.spark:spark-cassandra-connector\_2.12:3.5.0 spark/stream\_processor.py

Meanwhile, in terminal 2:

```
>activate taxic
>cd "C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream"
>python producer/producer.py
```

Now, opening the local host link : <http://localhost:4040/jobs/> , the Spark UI will appear.



| Job ID (Job Group)                        | Description                                                                                                                                  | Submitted           | Duration | Stages: Succeeded/Total | Tasks (for all stages): Succeeded/Total |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------|-------------------------|-----------------------------------------|
| 10 (7b0e4d4d-fc4b-4711-4729-c179b0a99993) | id = 0b04d2a-8d5d-4a4d-8a75-f1b6a8d777e runid = 7b0e4d4d-fc4b-4711-4729-c179b0a99993 batch = 21 start at hdfs://hdfs01.hadoopcloud.com/jobs/ | 2020/03/17 03:58:10 | 10 ms    | 1/1                     | 1/1                                     |
| 9 (81a7813-432a-4007-a88a-e877d8b78950)   | id = 5a64a4d-8d8f-4b5a-8716-3a71a1b3d8d7 runid = 81a7813-432a-4007-a88a-e877d8b78950 batch = 14 start at hdfs://hdfs01.hadoopcloud.com/jobs/ | 2020/03/17 03:58:08 | 2 s      | 2/2                     | 2/2                                     |
| 8 (7b0e4d4d-fc4b-4711-4729-c179b0a99993)  | id = 0b04d2a-8d5d-4a4d-8a75-f1b6a8d777e runid = 7b0e4d4d-fc4b-4711-4729-c179b0a99993 batch = 20 start at hdfs://hdfs01.hadoopcloud.com/jobs/ | 2020/03/17 03:58:07 | 0.7 s    | 1/1                     | 1/1                                     |
| 7 (7b0e4d4d-fc4b-4711-4729-c179b0a99993)  | id = 0b04d2a-8d5d-4a4d-8a75-f1b6a8d777e runid = 7b0e4d4d-fc4b-4711-4729-c179b0a99993 batch = 19 start at hdfs://hdfs01.hadoopcloud.com/jobs/ | 2020/03/17 03:58:04 | 0.8 s    | 1/1                     | 1/1                                     |
| 6 (81a7813-432a-4007-a88a-e877d8b78950)   | id = 5a64a4d-8d8f-4b5a-8716-3a71a1b3d8d7 runid = 81a7813-432a-4007-a88a-e877d8b78950 batch = 13 start at hdfs://hdfs01.hadoopcloud.com/jobs/ | 2020/03/17 03:58:04 | 2 s      | 2/2                     | 2/2                                     |
| 5 (7b0e4d4d-fc4b-4711-4729-c179b0a99993)  | id = 0b04d2a-8d5d-4a4d-8a75-f1b6a8d777e runid = 7b0e4d4d-fc4b-4711-4729-c179b0a99993 batch = 18 start at hdfs://hdfs01.hadoopcloud.com/jobs/ | 2020/03/17 03:58:02 | 0.3 s    | 1/1                     | 1/1                                     |
| 4 (7b0e4d4d-fc4b-4711-4729-c179b0a99993)  | id = 0b04d2a-8d5d-4a4d-8a75-f1b6a8d777e runid = 7b0e4d4d-fc4b-4711-4729-c179b0a99993 batch = 17 start at hdfs://hdfs01.hadoopcloud.com/jobs/ | 2020/03/17 03:55:59 | 0.1 s    | 1/1                     | 1/1                                     |
| 3 (81a7813-432a-4007-a88a-e877d8b78950)   | id = 5a64a4d-8d8f-4b5a-8716-3a71a1b3d8d7 runid = 81a7813-432a-4007-a88a-e877d8b78950 batch = 12 start at hdfs://hdfs01.hadoopcloud.com/jobs/ | 2020/03/17 03:55:59 | 2 s      | 2/2                     | 2/2                                     |
| 2 (7b0e4d4d-fc4b-4711-4729-c179b0a99993)  | id = 0b04d2a-8d5d-4a4d-8a75-f1b6a8d777e runid = 7b0e4d4d-fc4b-4711-4729-c179b0a99993 batch = 16 start at hdfs://hdfs01.hadoopcloud.com/jobs/ | 2020/03/17 03:55:57 | 0.6 s    | 1/1                     | 1/1                                     |
| 1 (7b0e4d4d-fc4b-4711-4729-c179b0a99993)  | id = 0b04d2a-8d5d-4a4d-8a75-f1b6a8d777e runid = 7b0e4d4d-fc4b-4711-4729-c179b0a99993 batch = 15 start at hdfs://hdfs01.hadoopcloud.com/jobs/ | 2020/03/17 03:55:52 | 2 s      | 1/1                     | 1/1                                     |
| 0 (81a7813-432a-4007-a88a-e877d8b78950)   | id = 5a64a4d-8d8f-4b5a-8716-3a71a1b3d8d7 runid = 81a7813-432a-4007-a88a-e877d8b78950 batch = 11 start at hdfs://hdfs01.hadoopcloud.com/jobs/ | 2020/03/17 03:55:52 | 5 s      | 2/2                     | 2/2                                     |

However, notice that this is actually less than what came earlier, and the more we run the pipeline like this, the less it may become.

This is because:

->When you deleted checkpoints, or didn't have them, Spark had to rebuild everything:

- Create state store
- Initialize stream metadata
- Build checkpoint directories
- Start Kafka offsets tracking
- Create Cassandra connection
- Run initial micro-batches

All this creates extra initialization jobs.

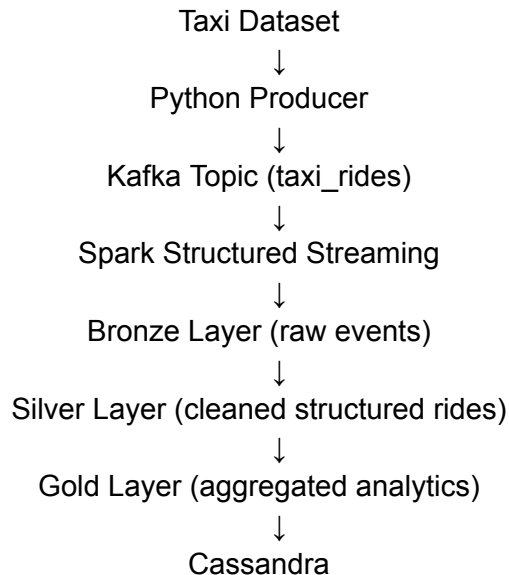
->->Meanwhile the second run had fewer jobs. This is because since we did not delete checkpoints, Spark already had:

- Kafka offsets
- State store metadata
- Query progress
- Schema information

---

### BRIEF OF WHAT WE HAVE DONE TILL NOW:

1. Built a real-time data engineering system:



2. Have completed the following steps:

- ✓ Streaming ingestion
- ✓ Schema parsing
- ✓ Medallion architecture (Bronze/Silver/Gold)
- ✓ Windowed aggregation
- ✓ Stateful streaming
- ✓ Cassandra sink
- ✓ Checkpointing
- ✓ Spark monitoring via UI

3. Recent Improvements of what we did in the cassandra stage of the project:

- window aggregation
- removal of unnecessary columns
- state store fix
- stability improvements

#### 4.COMMON ISSUES & RULES

1. If checkpoints are deleted → restart everything
2. If docker is down → Kafka topics are gone
3. If no data in UI → check producer
4. Silver has no checkpoint (not a streaming sink)
5. First run = many jobs, later runs = fewer jobs

---

### BRIEF OF WHAT HAS TO BE DONE NOW:

1. Real-time dashboard visualization using grafana, where dashboard metrics are:

- rides per minute
- average fare
- average trip distance

2. Workflow Orchestration using Airflow, which can:

- start Kafka
- start Spark job
- run producer
- monitor pipeline

3. Adding Additional metrics like:

- rides\_by\_location
- rides\_by\_payment\_type

If wanted in the future. These will be separate tables built in cqlsh, like:

gold\_rides\_by\_location

Gold\_by\_payment\_type

---

### STEP XIII: IMPLEMENTATION OF SURGE, ANOMALY, AND HOTSPOT DETECTION

1. Problem:

Till now, we had:

- Proper Bronze → Silver → Gold flow
- Watermark + window aggregation
- Cassandra integration working
- Clean schema handling

And our gold\_df only did: avg\_fare, avg\_trip\_distance, ride\_count;

Which was kind of just descriptive analytics

What was missing was a real-time decision system

So now our goal is to make a real-time decision-making system for ride demand and pricing.

2. Goal:

-> Adding Surge Detection

-> Anomaly Detection

-> Hotspot Detection

All inside the Spark Streaming Layer, and persist results properly in Cassandra (Gold Layer).

#### 2.1) The Surge Detection:

Surge detection identifies high-demand periods in real-time using ride density in time windows, enabling dynamic pricing and supply optimization.

"If too many rides are happening in a short time → demand is high"

Ex: when(ride\_count > 5, "HIGH\_DEMAND")

Basically;

- Too many people booking rides
- Drivers may be fewer
- System needs to react

Companies may use this feature to:

- Increase prices (surge pricing)
- Send more drivers to that area
- Reduce waiting time
- Rush due to Office hours
- Sudden rush due to rain

## 2.2) The Anomaly Detection:

Anomaly detection flags unusual fare patterns in real-time, helping identify pricing errors or abnormal ride conditions.

“If fare suddenly becomes too high → something unusual is happening”

Ex: `when(avg_fare > 100, "ABNORMAL")`

Basically;

- Pricing bug
- Fraud / overcharging
- Unexpected traffic or route

Companies may use this feature to:

- Detect pricing errors
- Prevent fraud
- Alert system failures

## 2.3) The Hotspot Detection:

Hotspot detection identifies high-demand locations based on ride density, enabling better driver allocation and operational efficiency.

“Which locations have high ride activity?”

Ex: `groupBy(window, pulocationid)`

Basically;

- Areas with high demand
- Busy zones (airports, offices, events)

Companies may use this feature to:

- Identify high-demand zones
- Send drivers to those locations
- Optimize driver distribution and planning driver positioning
- Reduce idle time

NOTE:

Surge = “WHEN demand is high” (time-based)

Ex: The Code: `groupBy(window("pickup_time", "1 minute"))`

Asks: “In this time window, how many rides happened?”

So if the ride count between 10:00 and 10:01 is 8, then it has high demand

Hotspot = “WHERE demand is high” (location-based)

Ex: The Code: `groupBy(window("pickup_time", "5 minutes"), pulocationid)`

Asks: “Which location has the most rides?”

So if the ride of a specific location, like an Airport, is 20, then it's a hotspot.

3. Before we start, we need to understand what changes and updations were taken for this stage of the project. The docker-compose.yml file was not touched. Major changes were taken in:

-> The stream\_processor.py file

-> The producer.py file (basically we increased the count of df.count(), for testing more.)

-> The cqlsh, taxi\_streaming database (basically addition of the hotspot table, and updation of the gold table)

4. Steps taken:

-> Modified the gold\_df block of the stream\_processor.py file such that the system can now detect real-time demand surges and pricing anomalies.

-> We created a new hotspot\_df , and hotspot\_query layer.

-> We altered the existing gold table in cqlsh:

>ALTER TABLE taxi\_streaming.gold\_rides ADD surge\_flag text;

>ALTER TABLE taxi\_streaming.gold\_rides ADD fare\_anomaly text;

-> We created the new hotspot layer:

```
>CREATE TABLE taxi_streaming.hotspot_rides (
 agg_date date,
 agg_time timestamp,
 pulocationid int,
 ride_count bigint,
 hotspot_flag text,
 PRIMARY KEY (agg_date, agg_time, pulocationid)
) WITH CLUSTERING ORDER BY (agg_time DESC);
```

Now in Cassandra gold\_rides table, we have-> avg\_fare, ride\_count, surge\_flag, fare\_anomaly, window\_start, window\_end .

And hotspot\_rides include-> pulocationid, ride\_count, hotspot\_flag, indow\_start, window\_end, agg\_time, agg\_date .

We now have built a real-time system that detects demand surges, fare anomalies, and geographic hotspots using streaming data.

---

## STEP XIV : IMPLEMENTATION OF MINOR CHANGES AND UPDATES

1. In the file producer.py,

The Code line: df = df.head(250), is just for testing purposes,

For real-time mimic of data streaming, it should be updated later to:

```
while True:
```

```
 for index, row in df.iterrows():
```

2. In the stream\_processor.py file:

We update the Kafka Offset setting:

The previous code of line, that was: .option("startingOffsets", "earliest")

Had a problem , and that was that with every restart, it used to process ALL data. This was bad for "production" explanation.

So we replaced it with : .option("startingOffsets", "latest")

Now, this line of code reads only new data, and works correctly with checkpointing.

This ensures the system processes only new incoming data and avoids reprocessing old Kafka messages, making it suitable for production-level streaming systems.

3. The current code has the follow thresholds:

- Surge : ride\_count > 5
- Hotspot: ride\_count > 10
- Anomaly: avg\_fare > 100

Which is perfect to actually see flags with our dataset. Thresholds may vary according to different data. It can be changed according to the dataset and size of batches.

4. The line of code: .

```
>withColumn("pickup_time", to_timestamp("tpep_pickup_datetime"))
```

For both gold\_df and hotspot\_df, is changed to:

```
>withColumn("pickup_time", col("pickup_datetime"))
```

This results in a cleaner pipeline and avoids redundant parsing.

5. Querying in dashboards becomes easier when we have the actual time grouping info.

So the line of code:

```
>drop("window")
```

Is replaced by:

```
>.withColumn("window_start", col("window.start")) \
 .withColumn("window_end", col("window.end")) \
 .drop("window")
```

In both gold\_df and hotspot\_df; but we also need to make changes in the taxi\_streaming tables in cqlsh too for that:

So in cqlsh;

```
>ALTER TABLE taxi_streaming.gold_rides ADD window_start timestamp;
>ALTER TABLE taxi_streaming.gold_rides ADD window_end timestamp;
>ALTER TABLE taxi_streaming.hotspot_rides ADD window_start timestamp;
>ALTER TABLE taxi_streaming.hotspot_rides ADD window_end timestamp;
```

---

## STEP XV: ALL UPDATED CODES TILL NOW

1. docker-compose.yml file:

```
services:

 zookeeper:
 image: confluentinc/cp-zookeeper:7.5.0
 container_name: zookeeper
 environment:
 ZOOKEEPER_CLIENT_PORT: 2181
 ports:
 - "2181:2181"

 kafka:
```

```

 image: confluentinc/cp-kafka:7.5.0
 container_name: kafka
 depends_on:
 - zookeeper
 ports:
 - "9092:9092"
 environment:
 KAFKA_BROKER_ID: 1
 KAFKA_ZOOKEEPER_CONNECT: zookeeper:2181
 KAFKA_ADVERTISED_LISTENERS: PLAINTEXT://localhost:9092
 KAFKA_OFFSETS_TOPIC_REPLICATION_FACTOR: 1
 cassandra:
 image: cassandra:4.1
 container_name: cassandra
 ports:
 - "9042:9042"
 environment:
 - CASSANDRA_CLUSTER_NAME=TaxiCluster
 volumes:
 - cassandra_data:/var/lib/cassandra

volumes:
 cassandra_data:

```

## 2. producer.py file:

```

from kafka import KafkaProducer
import pandas as pd
import json
import time

Kafka connection
producer = KafkaProducer(
 bootstrap_servers='localhost:9092',
 value_serializer=lambda v: json.dumps(v).encode('utf-8')
)

Load dataset
df = pd.read_parquet('data/yellow_tripdata_2025-01.parquet')

```

```

Limit rows for testing
df = df.head(250)

topic = "taxi_rides"

for index, row in df.iterrows():

 data = row.to_dict()

 # Convert timestamps to string
 for key, value in data.items():
 if hasattr(value, "isoformat"):
 data[key] = value.isoformat()

 producer.send(topic, value=data)

 print(f"Sent ride {index}")

 time.sleep(0.5) # simulate real-time streaming

```

### 3. stream\_processor.py file:

```

from pyspark.sql import SparkSession
from pyspark.sql.functions import to_date
from pyspark.sql.functions import col, from_json, current_timestamp, expr,
avg, count, to_timestamp
from pyspark.sql.types import StructType, StringType, DoubleType,
IntegerType
from pyspark.sql.functions import window

----- Spark Session -----
spark = SparkSession.builder \
 .appName("TaxiStreamingProcessor") \
 .config("spark.sql.shuffle.partitions", "2") \
 .config("spark.cassandra.connection.host", "127.0.0.1") \
 .config("spark.driver.host", "127.0.0.1") \
 .config("spark.sql.streaming.stateStore.providerClass",

"org.apache.spark.sql.execution.streaming.state.HDFSBackedStateStoreProvid
er") \

```

```

 .getOrCreate()

spark.catalog.clearCache()
spark.sparkContext.setLogLevel("WARN")

----- Full schema (all fields from your Kafka JSON)

schema = StructType() \
 .add("VendorID", IntegerType()) \
 .add("tpep_pickup_datetime", StringType()) \
 .add("tpep_dropoff_datetime", StringType()) \
 .add("passenger_count", DoubleType()) \
 .add("trip_distance", DoubleType()) \
 .add("RatecodeID", DoubleType()) \
 .add("store_and_fwd_flag", StringType()) \
 .add("PULocationID", IntegerType()) \
 .add("DOLocationID", IntegerType()) \
 .add("payment_type", IntegerType()) \
 .add("fare_amount", DoubleType()) \
 .add("extra", DoubleType()) \
 .add("mta_tax", DoubleType()) \
 .add("tip_amount", DoubleType()) \
 .add("tolls_amount", DoubleType()) \
 .add("improvement_surcharge", DoubleType()) \
 .add("total_amount", DoubleType()) \
 .add("congestion_surcharge", DoubleType()) \
 .add("Airport_fee", DoubleType()) \
 .add("cbd_congestion_fee", DoubleType())

----- Read Kafka Stream -----
df = spark.readStream \
 .format("kafka") \
 .option("kafka.bootstrap.servers", "localhost:9092") \
 .option("subscribe", "taxi_rides") \
 .option("startingOffsets", "latest") \
 .load()

json_df = df.selectExpr("CAST(value AS STRING) AS raw_event") \
 .withColumn("ride_id", expr("uuid()"))

----- BRONZE -----

```

```

bronze_df = json_df.withColumn("created_at", current_timestamp())
bronze_query = bronze_df.writeStream \
 .foreachBatch(lambda batch_df, _: batch_df.write
 .format("org.apache.spark.sql.cassandra")
 .options(table="bronze_rides", keyspace="taxi_streaming")
 .mode("append")
 .save()
) \
 .option("checkpointLocation", "checkpoints/bronze") \
 .outputMode("append") \
 .start()

----- SILVER -----
silver_df = json_df.select(
 from_json(col("raw_event"), schema).alias("data"),
 col("ride_id")
).select("ride_id", "data.*")

silver_df = silver_df.select(
 col("ride_id"),
 col("VendorID").alias("vendorid"),
 col("tpep_pickup_datetime"),
 col("tpep_dropoff_datetime"),
 col("passenger_count"),
 col("trip_distance"),
 col("RatecodeID").alias("ratecodeid"),
 col("store_and_fwd_flag"),
 col("PULocationID").alias("pulocationid"),
 col("DOLocationID").alias("dolocationid"),
 col("payment_type"),
 col("fare_amount"),
 col("extra"),
 col("mta_tax"),
 col("tip_amount"),
 col("tolls_amount"),
 col("improvement_surcharge"),
 col("total_amount"),
 col("congestion_surcharge"),
 col("Airport_fee").alias("airport_fee"),
 col("cbd_congestion_fee"),

```

```

 to_timestamp("tpep_pickup_datetime").alias("pickup_datetime"),
 to_timestamp("tpep_dropoff_datetime").alias("dropoff_datetime")
)
 # ----- GOLD -----
from pyspark.sql.functions import when

gold_df = (
 silver_df
 .withColumn("pickup_time", col("pickup_datetime"))
 .withWatermark("pickup_time", "10 minutes")
 .groupBy(window("pickup_time", "1 minute"))
 .agg(
 avg("fare_amount").alias("avg_fare"),
 avg("trip_distance").alias("avg_trip_distance"),
 count("*").alias("ride_count")
)
 .withColumn(
 "surge_flag",
 when(col("ride_count") > 5, "HIGH_DEMAND").otherwise("NORMAL")
)
 .withColumn(
 "fare_anomaly",
 when(col("avg_fare") > 100, "ABNORMAL").otherwise("NORMAL")
)
 .withColumn("agg_time", current_timestamp())
 .withColumn("agg_date", to_date("agg_time"))
 .withColumn("window_start", col("window.start"))
 .withColumn("window_end", col("window.end"))
 .drop("window")
)

hotspot_df = (
 silver_df
 .withColumn("pickup_time", col("pickup_datetime"))
 .withWatermark("pickup_time", "10 minutes")
 .groupBy(
 window("pickup_time", "5 minutes"),
 col("pulocationid")
)
 .agg(

```

```

 count("*").alias("ride_count")
)
 .withColumn(
 "hotspot_flag",
 when(col("ride_count") > 10, "HOTSPOT").otherwise("NORMAL")
)
 .withColumn("agg_time", current_timestamp())
 .withColumn("agg_date", to_date("agg_time"))
 .withColumn("window_start", col("window.start"))
 .withColumn("window_end", col("window.end"))
 .drop("window")
)

gold_query = gold_df.writeStream \
 .foreachBatch(lambda batch_df, _: batch_df.write
 .format("org.apache.spark.sql.cassandra")
 .options(table="gold_rides", keyspace="taxi_streaming")
 .mode("append")
 .save()
) \
 .option("checkpointLocation", "checkpoints/gold") \
 .outputMode("update") \
 .start()

hotspot_query = hotspot_df.writeStream \
 .foreachBatch(lambda batch_df, _: batch_df.write
 .format("org.apache.spark.sql.cassandra")
 .options(table="hotspot_rides", keyspace="taxi_streaming")
 .mode("append")
 .save()
) \
 .option("checkpointLocation", "checkpoints/hotspot") \
 .outputMode("update") \
 .start()

----- Await Termination -----
spark.streams.awaitAnyTermination()

```

---

## BRIEF OF WHAT WE HAVE DONE TILL NOW

1. We have now created a system that can be used by ride-hailing platforms to detect demand surges, pricing anomalies, and location-based hotspots in real-time for dynamic pricing and driver allocation.

We have:

- ride\_count
- avg\_fare
- surge detection
- anomaly detection
- hotspot detection

2. How to run project for now:

### CASE I : WHEN THERE IS NO CHECKPOINT FOLDER YET:

We delete the checkpoint folder from the main project file taxistream, if it's already present.

1. We go to Task Manager, and end these tasks if they exist:

- Docker Desktop
- Docker Desktop Backend
- com.docker.backend

Then we start Docker Desktop again, just by clicking on the Docker Desktop icon.

2. Now, in Terminal 1, we run the follow commands:

```
>activate taxic
>cd "C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream"
>docker compose down
>docker compose up -d
>docker exec -it kafka kafka-topics --create --topic taxi_rides --bootstrap-server
localhost:9092 --partitions 1 --replication-factor 1
```

(to create a kafka topic, has to be done everytime the docker has been up, after down )

```
>docker exec -it kafka kafka-topics --list --bootstrap-server localhost:9092
```

(to verify whether docker is running or not)

```
>set PYSPARK_PYTHON=C:\Users\rajve\anaconda3\envs\taxic\python.exe
>spark-submit --packages
```

**org.apache.spark:spark-sql-kafka-0-10\_2.12:3.5.8,com.datastax.spark:spark-cassandra-connector\_2.12:3.5.0 spark/stream\_processor.py**

And then basically it'll go up to somewhere like:

```
26/03/16 23:19:58 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, 127.0.0.1, 62535, None)
26/03/16 23:19:58 INFO SharedState: Setting hive.metastore.warehouse.dir ('null') to the value of spark.sql.warehouse.dir.
26/03/16 23:19:58 INFO SharedState: Warehouse path is 'file:/C:/Users/rajve/Desktop/SkillIssues/Personal%20Projects/taxistream/spark-warehouse'.
26/03/16 23:20:03 WARN ResolveWriteToStream: spark.sql.adaptive.enabled is not supported in streaming DataFrames/Datasets and will be disabled.
26/03/16 23:20:04 WARN AdminClientConfig: These configurations '[key.deserializer, value.deserializer, enable.auto.commit, max.poll.records, auto.offset.reset]' were supplied but are not used yet.
26/03/16 23:20:04 WARN ResolveWriteToStream: spark.sql.adaptive.enabled is not supported in streaming DataFrames/Datasets and will be disabled.
26/03/16 23:20:04 WARN AdminClientConfig: These configurations '[key.deserializer, value.deserializer, enable.auto.commit, max.poll.records, auto.offset.reset]' were supplied but are not used yet.
```

.....

4. Meanwhile in Terminal 2:

```
>activate taxic
```

```
>cd "C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream"
>python producer/producer.py
```

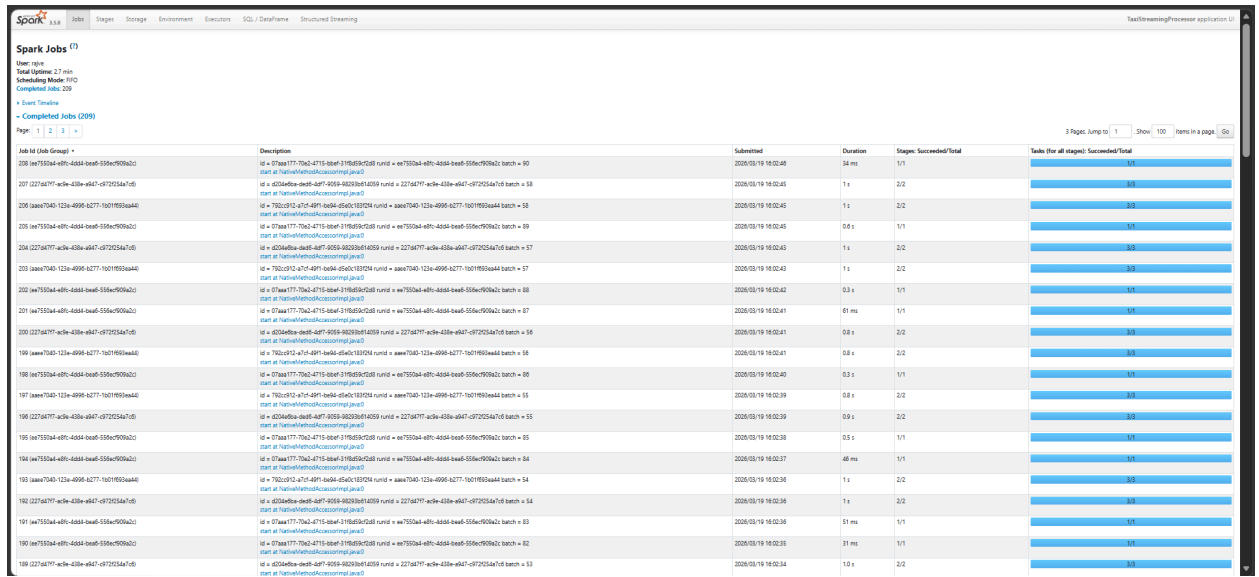
It'll send rides, this time till 250, since we set that. Later on we should put a while loop for continuous streaming of data from the dataset.

.....

To Verify, We go to local host:

http://localhost:4040/jobs/ ( [TaxiStreamingProcessor - Spark Jobs](#) )

Here, we'll be able to see the Spark UI:



| Job ID (Job Group)                        | Description                                                                                                                                 | Submitted           | Duration | Stages | Successful/Total |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------|--------|------------------|
| 208 (aa7535a4-4b7c-4d54-ba6b-555e909a2c)  | id = 07aaa177-7b62-4716-b0b7-31f5854c5f58 runid = aa7535a4-4b7c-4d54-ba6b-555e909a2c batch = 90<br>start at HadoopFileOutputCommitter.java# | 2020/03/19 16:02:48 | 34 ms    | 1/1    | 1/1              |
| 207 (2273a77f-ae3b-433e-a6d7-c97253a7a7d) | id = d23a6ba-d6d7-4058-8020b414359 runid = 2273a77f-ae3b-433e-a6d7-c97253a7a7d batch = 53<br>start at HadoopFileOutputCommitter.java#       | 2020/03/19 16:02:45 | 1 s      | 2/2    | 2/2              |
| 206 (aaa754d-123a-4999-4277-1b01f693a44)  | id = 762c932-27d-4d91-ba6a-d5a5c182f04 runid = aaa754d-123a-4999-4277-1b01f693a44 batch = 58<br>start at HadoopFileOutputCommitter.java#    | 2020/03/19 16:02:45 | 1 s      | 2/2    | 2/2              |
| 205 (aa7535a4-4b7c-4d54-ba6b-555e909a2c)  | id = 07aaa177-7b62-4716-b0b7-31f5854c5f58 runid = aa7535a4-4b7c-4d54-ba6b-555e909a2c batch = 69<br>start at HadoopFileOutputCommitter.java# | 2020/03/19 16:02:45 | 0.0 s    | 1/1    | 1/1              |
| 204 (2273a77f-ae3b-433e-a6d7-c97253a7a7d) | id = d23a6ba-d6d7-4058-8020b414359 runid = 2273a77f-ae3b-433e-a6d7-c97253a7a7d batch = 57<br>start at HadoopFileOutputCommitter.java#       | 2020/03/19 16:02:43 | 1 s      | 2/2    | 2/2              |
| 203 (aaa754d-123a-4999-4277-1b01f693a44)  | id = 762c932-27d-4d91-ba6a-d5a5c182f04 runid = aaa754d-123a-4999-4277-1b01f693a44 batch = 57<br>start at HadoopFileOutputCommitter.java#    | 2020/03/19 16:02:43 | 1 s      | 2/2    | 2/2              |
| 202 (aa7535a4-4b7c-4d54-ba6b-555e909a2c)  | id = 07aaa177-7b62-4716-b0b7-31f5854c5f58 runid = aa7535a4-4b7c-4d54-ba6b-555e909a2c batch = 68<br>start at HadoopFileOutputCommitter.java# | 2020/03/19 16:02:42 | 0.3 s    | 1/1    | 1/1              |
| 201 (aa7535a4-4b7c-4d54-ba6b-555e909a2c)  | id = 07aaa177-7b62-4716-b0b7-31f5854c5f58 runid = aa7535a4-4b7c-4d54-ba6b-555e909a2c batch = 67<br>start at HadoopFileOutputCommitter.java# | 2020/03/19 16:02:41 | 0.1 ms   | 1/1    | 1/1              |
| 200 (2273a77f-ae3b-433e-a6d7-c97253a7a7d) | id = d23a6ba-d6d7-4058-8020b414359 runid = 2273a77f-ae3b-433e-a6d7-c97253a7a7d batch = 56<br>start at HadoopFileOutputCommitter.java#       | 2020/03/19 16:02:41 | 0.0 s    | 2/2    | 2/2              |
| 199 (aaa754d-123a-4999-4277-1b01f693a44)  | id = 762c932-27d-4d91-ba6a-d5a5c182f04 runid = aaa754d-123a-4999-4277-1b01f693a44 batch = 56<br>start at HadoopFileOutputCommitter.java#    | 2020/03/19 16:02:41 | 0.0 s    | 2/2    | 2/2              |
| 198 (aa7535a4-4b7c-4d54-ba6b-555e909a2c)  | id = 07aaa177-7b62-4716-b0b7-31f5854c5f58 runid = aa7535a4-4b7c-4d54-ba6b-555e909a2c batch = 65<br>start at HadoopFileOutputCommitter.java# | 2020/03/19 16:02:40 | 0.3 s    | 1/1    | 1/1              |
| 197 (aaa754d-123a-4999-4277-1b01f693a44)  | id = 762c932-27d-4d91-ba6a-d5a5c182f04 runid = aaa754d-123a-4999-4277-1b01f693a44 batch = 55<br>start at HadoopFileOutputCommitter.java#    | 2020/03/19 16:02:39 | 0.0 s    | 2/2    | 2/2              |
| 196 (2273a77f-ae3b-433e-a6d7-c97253a7a7d) | id = d23a6ba-d6d7-4058-8020b414359 runid = 2273a77f-ae3b-433e-a6d7-c97253a7a7d batch = 55<br>start at HadoopFileOutputCommitter.java#       | 2020/03/19 16:02:39 | 0.0 s    | 2/2    | 2/2              |
| 195 (aa7535a4-4b7c-4d54-ba6b-555e909a2c)  | id = 07aaa177-7b62-4716-b0b7-31f5854c5f58 runid = aa7535a4-4b7c-4d54-ba6b-555e909a2c batch = 65<br>start at HadoopFileOutputCommitter.java# | 2020/03/19 16:02:38 | 0.5 s    | 1/1    | 1/1              |
| 194 (aa7535a4-4b7c-4d54-ba6b-555e909a2c)  | id = 07aaa177-7b62-4716-b0b7-31f5854c5f58 runid = aa7535a4-4b7c-4d54-ba6b-555e909a2c batch = 64<br>start at HadoopFileOutputCommitter.java# | 2020/03/19 16:02:37 | 48 ms    | 1/1    | 1/1              |
| 193 (aaa754d-123a-4999-4277-1b01f693a44)  | id = 762c932-27d-4d91-ba6a-d5a5c182f04 runid = aaa754d-123a-4999-4277-1b01f693a44 batch = 54<br>start at HadoopFileOutputCommitter.java#    | 2020/03/19 16:02:36 | 1 s      | 2/2    | 2/2              |
| 192 (2273a77f-ae3b-433e-a6d7-c97253a7a7d) | id = d23a6ba-d6d7-4058-8020b414359 runid = 2273a77f-ae3b-433e-a6d7-c97253a7a7d batch = 54<br>start at HadoopFileOutputCommitter.java#       | 2020/03/19 16:02:36 | 1 s      | 2/2    | 2/2              |
| 191 (aa7535a4-4b7c-4d54-ba6b-555e909a2c)  | id = 07aaa177-7b62-4716-b0b7-31f5854c5f58 runid = aa7535a4-4b7c-4d54-ba6b-555e909a2c batch = 63<br>start at HadoopFileOutputCommitter.java# | 2020/03/19 16:02:36 | 51 ms    | 1/1    | 1/1              |
| 190 (aa7535a4-4b7c-4d54-ba6b-555e909a2c)  | id = 07aaa177-7b62-4716-b0b7-31f5854c5f58 runid = aa7535a4-4b7c-4d54-ba6b-555e909a2c batch = 62<br>start at HadoopFileOutputCommitter.java# | 2020/03/19 16:02:35 | 31 ms    | 1/1    | 1/1              |
| 189 (2273a77f-ae3b-433e-a6d7-c97253a7a7d) | id = d23a6ba-d6d7-4058-8020b414359 runid = 2273a77f-ae3b-433e-a6d7-c97253a7a7d batch = 53<br>start at HadoopFileOutputCommitter.java#       | 2020/03/19 16:02:34 | 1.0 s    | 2/2    | 2/2              |

For 250 rides sent, 209 jobs were completed.

This good as one batch of work can have more than 1 ride, or sometimes 2 or 3 too.

We can also check whether Cassandra is giving results or not, by going to the cqlsh shell. This is also to be done in the terminal 2, as terminal 1 is still running spark.

```
>docker exec -t cassandra cqlsh
```

```
cqlsh> SELECT ride_count, surge_flag, fare_anomaly, window_start FROM
taxi_streaming.gold_rides LIMIT 10;
```

```
cqlsh> SELECT pulocationid, ride_count, hotspot_flag, window_start FROM
taxi_streaming.hotspot_rides LIMIT 10;
```

```
cqlsh> SELECT window_start, window_end FROM taxi_streaming.gold_rides LIMIT 5;
```

```
cqlsh> SELECT ride_count, surge_flag, fare_anomaly, window_start FROM taxi_streaming.gold_rides LIMIT 10;
```

| ride_count | surge_flag  | fare_anomaly | window_start                    |
|------------|-------------|--------------|---------------------------------|
| 3          |             | NORMAL       | 2024-12-31 19:28:00.000000+0000 |
| 6          | HIGH_DEMAND | NORMAL       | 2024-12-31 19:23:00.000000+0000 |
| 4          |             | NORMAL       | 2024-12-31 19:24:00.000000+0000 |
| 5          |             | NORMAL       | 2024-12-31 19:23:00.000000+0000 |
| 2          |             | NORMAL       | 2024-12-31 19:28:00.000000+0000 |
| 1          |             | NORMAL       | 2024-12-31 19:31:00.000000+0000 |
| 3          |             | NORMAL       | 2024-12-31 19:24:00.000000+0000 |
| 4          |             | NORMAL       | 2024-12-31 19:23:00.000000+0000 |
| 4          |             | NORMAL       | 2024-12-31 19:22:00.000000+0000 |
| 1          |             | NORMAL       | 2024-12-31 19:28:00.000000+0000 |

(10 rows)

```
cqlsh> SELECT pulocationid, ride_count, hotspot_flag, window_start FROM taxi_streaming.hotspot_rides LIMIT 10;
```

| pulocationid | ride_count | hotspot_flag | window_start                    |
|--------------|------------|--------------|---------------------------------|
| 68           | 1          | NORMAL       | 2024-12-31 19:25:00.000000+0000 |
| 107          | 1          | NORMAL       | 2024-12-31 19:20:00.000000+0000 |
| 142          | 2          | NORMAL       | 2024-12-31 19:20:00.000000+0000 |
| 237          | 3          | NORMAL       | 2024-12-31 19:20:00.000000+0000 |
| 263          | 1          | NORMAL       | 2024-12-31 19:25:00.000000+0000 |
| 261          | 1          | NORMAL       | 2024-12-31 19:30:00.000000+0000 |
| 229          | 2          | NORMAL       | 2024-12-31 19:20:00.000000+0000 |
| 229          | 1          | NORMAL       | 2024-12-31 19:20:00.000000+0000 |
| 263          | 2          | NORMAL       | 2024-12-31 19:20:00.000000+0000 |
| 249          | 2          | NORMAL       | 2024-12-31 19:20:00.000000+0000 |

```
cqlsh> SELECT window_start, window_end FROM taxi_streaming.gold_rides LIMIT 5;
```

| window_start                    | window_end                      |
|---------------------------------|---------------------------------|
| 2024-12-31 19:28:00.000000+0000 | 2024-12-31 19:29:00.000000+0000 |
| 2024-12-31 19:23:00.000000+0000 | 2024-12-31 19:24:00.000000+0000 |
| 2024-12-31 19:24:00.000000+0000 | 2024-12-31 19:25:00.000000+0000 |
| 2024-12-31 19:23:00.000000+0000 | 2024-12-31 19:24:00.000000+0000 |
| 2024-12-31 19:28:00.000000+0000 | 2024-12-31 19:29:00.000000+0000 |

(5 rows)

```
cqlsh> SELECT window_start, ride_count FROM taxi_streaming.gold_rides WHERE surge_flag = 'HIGH_DEMAND' ALLOW FILTERING;
```

```
cqlsh> SELECT window_start, ride_count FROM taxi_streaming.gold_rides WHERE surge_flag = 'HIGH_DEMAND' ALLOW FILTERING;
```

| window_start                    | ride_count |
|---------------------------------|------------|
| 2024-12-31 19:23:00.000000+0000 | 6          |
| 2024-12-31 19:25:00.000000+0000 | 7          |
| 2024-12-31 19:25:00.000000+0000 | 7          |
| 2024-12-31 19:25:00.000000+0000 | 6          |

```
cqlsh> SELECT * FROM taxi_streaming.gold_rides LIMIT 5;
```

```
cqlsh> SELECT * FROM taxi_streaming.gold_rides LIMIT 5;
```

| agg_date<br>ow_end | agg_time            | window_start        | avg_fare    | avg_trip_distance | fare_anomaly | ride_count | surge_flag  | wind |
|--------------------|---------------------|---------------------|-------------|-------------------|--------------|------------|-------------|------|
| 2026-03-19         | 2026-03-19 10:32:40 | 438000+0000         | 16.06667    | 2.16              | NORMAL       | 3          | NORMAL      | 2024 |
| -12-31 19:29:00    | 000000+0000         | 2024-12-31 19:28:00 | 000000+0000 | 1.59667           | NORMAL       | 6          | HIGH_DEMAND | 2024 |
| 2026-03-19         | 2026-03-19 10:32:38 | 350000+0000         | 11.4        | 1.7               | NORMAL       | 4          | NORMAL      | 2024 |
| -12-31 19:24:00    | 000000+0000         | 2024-12-31 19:23:00 | 000000+0000 | 1.628             | NORMAL       | 5          | NORMAL      | 2024 |
| 2026-03-19         | 2026-03-19 10:32:32 | 139000+0000         | 14.9        | 1.87              | NORMAL       | 2          | NORMAL      | 2024 |
| -12-31 19:25:00    | 000000+0000         | 2024-12-31 19:24:00 | 000000+0000 |                   |              |            |             |      |
| 2026-03-19         | 2026-03-19 10:32:24 | 030000+0000         | 11.26       |                   |              |            |             |      |
| -12-31 19:24:00    | 000000+0000         | 2024-12-31 19:23:00 | 000000+0000 |                   |              |            |             |      |
| 2026-03-19         | 2026-03-19 10:32:19 | 880000+0000         | 13.15       |                   |              |            |             |      |
| -12-31 19:29:00    | 000000+0000         | 2024-12-31 19:28:00 | 000000+0000 |                   |              |            |             |      |

(5 rows)

If ever asked: Why are we using ‘ALOOOW FILTERING’

We say: Cassandra is optimized for query-based modeling. Since this is an exploratory/demo setup, I used ALLOW FILTERING. In production, I would redesign the table based on query patterns.

## CASE II : WHEN THE CHECKPOINT FOLDER IS ALREADY THERE, AND DOCKER WAS NEVER DONE DOWN:

If docker wasn’t done down and the checkpoints were not deleted.

In terminal 1:

```
>activate taxic
>cd "C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream"
>set PYSARK_PYTHON=C:\Users\rajve\anaconda3\envs\taxic\python.exe
>spark-submit --packages
```

org.apache.spark:spark-sql-kafka-0-10\_2.12:3.5.8,com.datastax.spark:spark-cassandra-connector\_2.12:3.5.0 spark/stream\_processor.py

Meanwhile, in terminal 2:

```
>activate taxic
>cd "C:\Users\rajve\Desktop\SkillIssues\Personal Projects\taxistream"
>python producer/producer.py
```

Now, opening the local host link :<http://localhost:4040/jobs/> , the Spark UI will appear.

| Spark 3.5.8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |         |             |           |                 |                      | Telemetry/Transect application |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------|-------------|-----------|-----------------|----------------------|--------------------------------|--|
| Info                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Logs | Storage | Environment | Executors | SQL / DataFrame | Structured Streaming |                                |  |
| <b>Spark Jobs</b> <sup>(7)</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |      |         |             |           |                 |                      |                                |  |
| User: rajve                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |         |             |           |                 |                      |                                |  |
| Total Uptime: 3.0 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |         |             |           |                 |                      |                                |  |
| Scheduling Mode: FIFO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |         |             |           |                 |                      |                                |  |
| Completed Jobs: 170                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |         |             |           |                 |                      |                                |  |
| + Exec Timeline                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |         |             |           |                 |                      |                                |  |
| Completed Jobs (170)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |      |         |             |           |                 |                      |                                |  |
| Page: 1   2   3   4   5   6   7   8   9   10   11   12   13   14   15   16   17   18   19   20   21   22   23   24   25   26   27   28   29   30   31   32   33   34   35   36   37   38   39   40   41   42   43   44   45   46   47   48   49   50   51   52   53   54   55   56   57   58   59   60   61   62   63   64   65   66   67   68   69   70   71   72   73   74   75   76   77   78   79   80   81   82   83   84   85   86   87   88   89   90   91   92   93   94   95   96   97   98   99   100   101   102   103   104   105   106   107   108   109   110   111   112   113   114   115   116   117   118   119   120   121   122   123   124   125   126   127   128   129   130   131   132   133   134   135   136   137   138   139   140   141   142   143   144   145   146   147   148   149   150   151   152   153   154   155   156   157   158   159   160   161   162   163   164   165   166   167   168   169   170   171   172   173   174   175   176   177   178   179   180   181   182   183   184   185   186   187   188   189   190   191   192   193   194   195   196   197   198   199   200   201   202   203   204   205   206   207   208   209   210   211   212   213   214   215   216   217   218   219   220   221   222   223   224   225   226   227   228   229   230   231   232   233   234   235   236   237   238   239   240   241   242   243   244   245   246   247   248   249   250   251   252   253   254   255   256   257   258   259   260   261   262   263   264   265   266   267   268   269   270   271   272   273   274   275   276   277   278   279   280   281   282   283   284   285   286   287   288   289   290   291   292   293   294   295   296   297   298   299   300   301   302   303   304   305   306   307   308   309   310   311   312   313   314   315   316   317   318   319   320   321   322   323   324   325   326   327   328   329   330   331   332   333   334   335   336   337   338   339   340   341   342   343   344   345   346   347   348   349   350   351   352   353   354   355   356   357   358   359   360   361   362   363   364   365   366   367   368   369   370   371   372   373   374   375   376   377   378   379   380   381   382   383   384   385   386   387   388   389   390   391   392   393   394   395   396   397   398   399   400   401   402   403   404   405   406   407   408   409   410   411   412   413   414   415   416   417   418   419   420   421   422   423   424   425   426   427   428   429   430   431   432   433   434   435   436   437   438   439   440   441   442   443   444   445   446   447   448   449   450   451   452   453   454   455   456   457   458   459   460   461   462   463   464   465   466   467   468   469   470   471   472   473   474   475   476   477   478   479   480   481   482   483   484   485   486   487   488   489   490   491   492   493   494   495   496   497   498   499   500   501   502   503   504   505   506   507   508   509   510   511   512   513   514   515   516   517   518   519   520   521   522   523   524   525   526   527   528   529   530   531   532   533   534   535   536   537   538   539   540   541   542   543   544   545   546   547   548   549   550   551   552   553   554   555   556   557   558   559   560   561   562   563   564   565   566   567   568   569   570   571   572   573   574   575   576   577   578   579   580   581   582   583   584   585   586   587   588   589   590   591   592   593   594   595   596   597   598   599   600   601   602   603   604   605   606   607   608   609   610   611   612   613   614   615   616   617   618   619   620   621   622   623   624   625   626   627   628   629   630   631   632   633   634   635   636   637   638   639   640   641   642   643   644   645   646   647   648   649   650   651   652   653   654   655   656   657   658   659   660   661   662   663   664   665   666   667   668   669   670   671   672   673   674   675   676   677   678   679   680   681   682   683   684   685   686   687   688   689   690   691   692   693   694   695   696   697   698   699   700   701   702   703   704   705   706   707   708   709   710   711   712   713   714   715   716   717   718   719   720   721   722   723   724   725   726   727   728   729   730   731   732   733   734   735   736   737   738   739   740   741   742   743   744   745   746   747   748   749   750   751   752   753   754   755   756   757   758   759   760   761   762   763   764   765   766   767   768   769   770   771   772   773   774   775   776   777   778   779   780   781   782   783   784   785   786   787   788   789   790   791   792   793   794   795   796   797   798   799   800   801   802   803   804   805   806   807   808   809   810   811   812   813   814   815   816   817   818   819   820   821   822   823   824   825   826   827   828   829   830   831   832   833   834   835   836   837   838   839   840   841   842   843   844   845   846   847   848   849   850   851   852   853   854   855   856   857   858   859   860   861   862   863   864   865   866   867   868   869   870   871   872   873   874   875   876   877   878   879   880   881   882   883   884   885   886   887   888   889   890   891   892   893   894   895   896   897   898   899   900   901   902   903   904   905   906   907   908   909   910   911   912   913   914   915   916   917   918   919   920   921   922   923   924   925   926   927   928   929   930   931   932   933   934   935   936   937   938   939   940   941   942   943   944   945   946   947   948   949   950   951   952   953   954   955   956   957   958   959   960   961   962   963   964   965   966   967   968   969   970   971   972   973   974   975   976   977   978   979   980   981   982   983   984   985   986   987   988   989   990   991   992   993   994   995   996   997   998   999   1000   1001   1002   1003   1004   1005   1006   1007   1008   1009   1010   1011   1012   1013   1014   1015   1016   1017   1018   1019   1020   1021   1022   1023   1024   1025   1026   1027   1028   1029   1030   1031   1032   1033   1034   1035   1036   1037   1038   1039   1040   1041   1042   1043   1044   1045   1046   1047   1048   1049   1050   1051   1052   1053   1054   1055   1056   1057   1058   1059   1060   1061   1062   1063   1064   1065   1066   1067   1068   1069   1070   1071   1072   1073   1074   1075   1076   1077   1078   1079   1080   1081   1082   1083   1084   1085   1086   1087   1088   1089   1090   1091   1092   1093   1094   1095   1096   1097   1098   1099   1100   1101   1102   1103   1104   1105   1106   1107   1108   1109   1110   1111   1112   1113   1114   1115   1116   1117   1118   1119   1120   1121   1122   1123   1124   1125   1126   1127   1128   1129   1130   1131   1132   1133   1134   1135   1136   1137   1138   1139   1140   1141   1142   1143   1144   1145   1146   1147   1148   1149   1150   1151   1152   1153   1154   1155   1156   1157   1158   1159   1160   1161   1162   1163   1164   1165   1166   1167   1168   1169   1170   1171   1172   1173   1174   1175   1176   1177   1178   1179   1180   1181   1182   1183   1184   1185   1186   1187   1188   1189   1190   1191   1192   1193   1194   1195   1196   1197   1198   1199   1200   1201   1202   1203   1204   1205   1206   1207   1208   1209   1210   1211   1212   1213   1214   1215   1216   1217   1218   1219   1220   1221   1222   1223   1224   1225   1226   1227   1228   1229   1230   1231   1232   1233   1234   1235   1236   1237   1238   1239   1240   1241   1242   1243   1244   1245   1246   1247   1248   1249   1250   1251   1252   1253   1254   1255   1256   1257   1258   1259   1260   1261   1262   1263   1264   1265   1266   1267   1268   1269   1270   1271   1272   1273   1274   1275   1276   1277   1278   1279   1280   1281   1282   1283   1284   1285   1286   1287   1288   1289   1290   1291   1292   1293   1294   1295   1296   1297   1298   1299   1300   1301   1302   1303   1304   1305   1306   1307   1308   1309   1310   1311   1312   1313   1314   1315   1316   1317   1318   1319   1320   1321   1322   1323   1324   1325   1326   1327   1328   1329   1330   1331   1332   1333   1334   1335   1336   1337   1338   1339   1340   1341   1342   1343   1344   1345   1346   1347   1348   1349   1350   1351   1352   1353   1354   1355   1356   1357   1358   1359   1360   1361   1362   1363   1364   1365   1366   1367   1368   1369   1370   1371   1372   1373   1374   1375   1376   1377   1378   1379   1380   1381   1382   1383   1384   1385   1386   1387   1388   1389   1390   1391   1392   1393   1394   1395   1396   1397   1398   1399   1400   1401   1402   1403   1404   1405   1406   1407   1408   1409   1410   1411   1412   1413   1414   1415   1416   1417   1418   1419   1420   1421   1422   1423   1424   1425   1426   1427   1428   1429   1430   1431   1432   1433   1434   1435   1436   1437   1438   1439   1440   1441   1442   1443   1444   1445   1446   1447   1448   1449   1450   1451   1452   1453   1454   1455   1456   1457   1458   1459   1460   1461   1462   1463   1464   1465   1466   1467   1468   1469   1470   1471   1472   1473   1474   1475   1476   1477   1478   1479   1480   1481   1482   1483   1484   1485   1486   1487   1488   1489   1490   1491   1492   1493   1494   1495   1496   1497   1498   1499   1500   1501   1502   1503   1504   1505   1506   1507   1508   1509   1510   1511   1512   1513   1514   1515   1516   1517   1518   1519   1520   1521   1522   1523   1524   1525   1526   1527   1528   1529   1530   1531   1532   1533   1534   1535   1536   1537   1538   1539   1540   1541   1542   1543   1544   1545   1546   1547   1548   1549   1550   1551   1552   1553   1554   1555   1556   1557   1558   1559   1560   1561   1562   1563   1564   1565   1566   1567   1568   1569   1570   1571   1572   1573   1574   1575   1576   1577   1578   1579   1580   1581   1582   1583   1584   1585   1586   1587   1588   1589   1590   1591   1592   1593   1594   1595   1596   1597   1598   1599   1600   1601   1602   1603   1604   1605   1606   1607   1608   1609   1610   1611   1612   1613   1614   1615   1616   1617   1618   1619   1620   1621   1622   1623   1624   1625   1626   1627   1628   1629   1630   1631   1632   1633   1634   1635   1636   1637   1638   1639   1640   1641   1642   1643   1644   1645   1646   1647   1648   1649   1650   1651   1652   1653   1654   1655   1656   1657   1658   1659   1660   1661   1662   1663   1664   1665   1666   1667   1668   1669   1670   1671   1672   1673   1674   1675   1676   1677   1678   1679   1680   1681   1682   1683   1684   1685   1686   1687   1688   1689   1690   1691   1692   1693   1694   1695   1696   1697   1698   1699   1700   1701   1702   1703   1704   1705   1706   1707   1708   1709   1710   1711   1712   1713   1714   1715   1716   1717   1718   1719   1720   1721   1722   1723   1724   1725   1726   1727   1728   1729   1730   1731   1732   1733   1734   1735   1736   1737   1738   1739   1740   1741   1742   1743   1744   1745   1746   1747   1748   1749   1750   1751   1752   1753   1754   1755   1756   1757   1758   1759   1760   1761   1762   1763   1764   1765   1766   1767   1768   1769   1770   1771   1772   1773   1774   1775   1776   1777   1778   1779   1780   1781   1782   1783   1784   1785   1786   1787   1788   1789   1790   1791   1792   1793   1794   1795   1796   1797   1798   1799   1800   1801   1802   1803   1804   1805   1806   1807   1808   1809   1810   1811   1812   1813   1814   1815   1816   1817   1818   1819   1820   1821   1822   1823   1824   1825   1826   1827   1828   1829   1830   1831   1832   1833   1834   1835   1836   1837   1838   1839   1840   1841   1842   1843   1844   1845   1846   1847   1848   1849   1850   1851   1852   1853   1854   1855   1856   1857   1858   1859   1860   1861   1862   1863   1864   1865   1866   1867   1868   1869   1870   1871   1872   1873   1874   1875   1876   1877   1878   1879   1880   1881   1882   1883   1884   1885   1886   1887   1888   1889   1890   1891   1892   1893   1894   1895   1896   1897   1898   1899   1900   1901   1902   1903   1904   1905   1906   1907   1908   1909   1910   1911   1912   1913   1914   1915   1916   1917   1918   1919   1920   1921   1922   1923   1924   1925   1926   1927   1928   1929   1930   1931   1932   1933   1934   1935   1936   1937   1938   1939   1940   1941   1942   1943   1944   1945   1946   1947   1948   1949   1950   1951   1952   1953   1954   1955   1956   1957   1958   1959   1960   1961   1962   1963   1964   1965   1966   1967   1968   1969   1970   1971   1972   1973   1974   1975   1976   1977   1978   1979   1980   1981   1982   1983   1984   1985   1986   1987   1988   1989   1990   1991   1992   1993   1994   1995   1996   1997   1998   1999   2000   2001   2002   2003   2004   2005   2006   2007   2008   2009   2010   2011   2012   2013   2014   2015   2016   2017   2018   2019   2020   2021   2022   2023   2024   2025   2026   2027   2028   2029   2030   2031   2032   2033   2034   2035   2036   2037   2038   2039   2040   2041   2042   2043   2044   2045   2046   2047   2048   2049   2050   2051   2052   2053   2054   2055   2056   2057   2058   2059   2060   2061   2062   2063   2064   2065   2066   2067   2068   2069   2070   2071   2072   2073   2074   2075   2076   2077   2078   2079   2080   2081   2082   2083   2084   2085   2086   2087   2088   2089   2090   2091   2092   2093   2094   2095   2096   2097   2098   2099   2100   2101   2102   2103   2104   2105   2106   2107   2108   2109   2110   2111   2112   2113   2114   2115   2116   2117   2118   2119   2120   2121   2122   2123   2124   2125   2126   2127   2128   2129   2130   2131   2132   2133   2134   2135   2136   2137   2138   2139   2140   2141   2142   2143   2144   2145   2146   2147   2148   2149   2150   2151   2152   2153   2154   2155   2156   2157   2158   2159   2160   2161   2162   2163   2164   2165   2166   2167   2168   2169   2170   2171   2172   2173   2174   2175   2176   2177   2178   2179   2180   2181   2182   2183   2184   2185   2186   2187   2188   2189   2190   2191   2192   2193   2194   2195   2196   2197   2198   2199   2200   2201   2202   2203   2204   2205   2206   2207   2208   2209   22 |      |         |             |           |                 |                      |                                |  |

-> We notice that now jobs are 170, this is because Spark is resuming from checkpoints and Kafka offsets, not reprocessing everything again.

-> Spark Structured Streaming uses checkpointing and Kafka offset tracking to ensure fault tolerance and exactly-once processing. On restart, it resumes from the last processed offset instead of reprocessing the entire stream, resulting in fewer jobs.

-> Even if the same dataset is sent again, Kafka treats each record as a new event with a new offset. Hence, Spark processes it as new streaming data.

-> But then the question may arise: Why even 170, why not 0? Because literally we gave in the same 250 rides before, so shouldn't it already have been in the checkpoints?

-> The answer is that Spark doesn't track "records". It tracks offsets + state + time windows.

-> So when we ran the producer.py file, Even though it's the same dataset, this line:

```
>producer.send(topic, value=data) ; sends new messages to Kafka.
```

-> Kafka sees this as: NEW MESSAGE → NEW OFFSET

NOT: same message → skip

-> The second reason for this is the: Checkpoint state reuse

Spark already knows:

- Window aggregations
- Partial computations
- Previous state

So, it processes faster + fewer batches

-> The third reason is due to Micro-batch timing differences. The `time.sleep(0.5)`, is never identical. So batch grouping may change slightly, and the number of jobs may change.

---

### **BRIEF OF WHAT HAS TO BE DONE NOW:**

#### **1. Grafana Dashboard**

- Connect Grafana to Cassandra
- Visualize:
  - ride\_count over time
  - surge\_flag trends
  - hotspot locations
  - avg\_fare trends
  - Create meaningful panels
  - average trip distance

#### **2. Airflow Integration**

- Schedule producer and Spark jobs
  - Manage pipeline execution
  - Create DAG
-